



SmarTyper Tutorial

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<https://github.com/rocpengliu/SmarTyper>

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Step 1 File preparation

Step 1.1 Folder containing fastq.gz files for each sample

 CMS031_AT_20230817

```
liup@WPNRN771507:~/project/data/prnp/newtest/CMS031_AT_20230817$ ls
20919_S302_L001_R1_001.fastq.gz  21299_S297_L001_R1_001.fastq.gz  22745_S348_L001_R1_001.fastq.gz
20919_S302_L001_R2_001.fastq.gz  21299_S297_L001_R2_001.fastq.gz  22745_S348_L001_R2_001.fastq.gz
21093_S277_L001_R1_001.fastq.gz  21301_S298_L001_R1_001.fastq.gz  22746_S349_L001_R1_001.fastq.gz
21093_S277_L001_R2_001.fastq.gz  21301_S298_L001_R2_001.fastq.gz  22746_S349_L001_R2_001.fastq.gz
21094_S278_L001_R1_001.fastq.gz  21303_S300_L001_R1_001.fastq.gz  22748_S350_L001_R1_001.fastq.gz
21094_S278_L001_R2_001.fastq.gz  21303_S300_L001_R2_001.fastq.gz  22748_S350_L001_R2_001.fastq.gz
21095_S279_L001_R1_001.fastq.gz  21304_S301_L001_R1_001.fastq.gz  22750_S351_L001_R1_001.fastq.gz
```

Step 1.2 Sample file consisting of 4 columns for PE reads or 3 columns for SE reads separated by 'tab'

 samplefile_test.txt

Col1: sample name

Col3: reverse fastq.gz file

```
20919_S302_L001 20919_S302_L001_R1_001.fastq.gz 20919_S302_L001_R2_001.fastq.gz gpt
21093_S277_L001 21093_S277_L001_R1_001.fastq.gz 21093_S277_L001_R2_001.fastq.gz gpt
21094_S278_L001 21094_S278_L001_R1_001.fastq.gz 21094_S278_L001_R2_001.fastq.gz gpt
21095_S279_L001 21095_S279_L001_R1_001.fastq.gz 21095_S279_L001_R2_001.fastq.gz gpt
21098_S280_L001 21098_S280_L001_R1_001.fastq.gz 21098_S280_L001_R2_001.fastq.gz gpt
21099_S281_L001 21099_S281_L001_R1_001.fastq.gz 21099_S281_L001_R2_001.fastq.gz gpt
21101_S282_L001 21101_S282_L001_R1_001.fastq.gz 21101_S282_L001_R2_001.fastq.gz gpt
21102_S283_L001 21102_S283_L001_R1_001.fastq.gz 21102_S283_L001_R2_001.fastq.gz gpt
21104_S284_L001 21104_S284_L001_R1_001.fastq.gz 21104_S284_L001_R2_001.fastq.gz gpt
21105_S285_L001 21105_S285_L001_R1_001.fastq.gz 21105_S285_L001_R2_001.fastq.gz gpt
```

Col2: forward fastq.gz file

Col4: group name

Step 1.3 Loci file consisting of 8 columns separated by 'tab'

 prnp_sudo_file.txt

Loci file: 8 columns separated by 'tab'



CGCTGGAGCAGTGGTAGG

0 6
GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAA
CTACAGGCCAGTGGATCAGTATAATAACCAGAACACCTTTGTGCATGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCAG
ACTGACATTAAGATGATGGAGCGAGTTGTGGAGCAAATGTGCATCACCCAGTACCAGAGAGAATCCCAGGCTTATTACCAAAGAGG
GTGATCCTCCTCATCTCT

PRNP

366
TTCCTCATTTTTTCTCATAGTAGGA

Exon: 1:(366+1)

Position starts at 0
(0-based index)

CGCTGGAGCAGTGGTAGG

GGGCCTCGGTGGC
3 12
TACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTACTACAGGCCAGTG
GATCAGTATAATAACCAGAACACCTTTGTGCATGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAAGTTACCGAAACTGACATTAAGAT
GATGGAGCGAGTTGTGGAGCAAATGTGCATCACCCAGTACCAGAGAGAATCCCAGGCTTATTACCAAAGAGGGGCAAGTGTGATCCTCTTCTCCTCCCTCCTGT

PRNP2

GATCCTCCTCATCTCT

Exon: 3:(335+1)

335

TTCCTCATTTTTTCTCATAGTAGGA

locus	fp	rp	trim left length	trim right length	target SNP position	cds pos (start:end)	ref seq
PRNP	CGCTGGAGCAGTGGTAGG	TTCCTCATTTTTTCTCATAGTAGGA	0	0	6 7 35 60 80 127 141 147 148 156 197 198 347 365 366	1:367	GGGCCTCGGTGGCTACATGCTGGGAAGTG
PRNP2	CGCTGGAGCAGTGGTAGG	TTCCTCATTTTTTCTCATAGTAGGA	3	10	12 17 30	3:150 180:336	TACATGCTGGGAAGTGCCATGAGCAGGCCT
F2	GGATGGCAGCCTTACTGGGC	GAATCGAGATGCCAGTGCCCTCTG	0	0	0	2:65	CCTGGTGCTACACCACATCCCCTACTCTGCC
14_16000806	AAAGTGCATATGGAACCAGATCAA	TCATTGCACACCTACTATTGGAAC	0	0	0	0	CTCCTTAACACAGAGACATTTTAAACAAAA
15_18072427	ACTACACGAACCAAGAACATAAAA	CCTTAAAGAAGTGGTAGAATCTGTG	0	0	0	0	GGCTCGAGGGTTGAGGACAACAAGTTCTGC
17_1114348	TTAGCAGGGTTGCACAAAGT	AGGGAATGGCATTGTTTCCA	0	0	0	0	GGGAGGTTGGTTATTTAATTTTCTATATG

If your target SNP is at position 0, your input is 0|

Step 2 Installation and launch SmarTyper

Getting started

Step 1. Pre-installations

SmarTyper is written in Python/Cython and C/C++11, and can be installed on Linux. We tested SmarTyper on Ubuntu 24.04.4 LTS. If you use Windows, installing and running SmarTyper through Windows Subsystem for Linux 2 (WSL2) is recommended.

Install system dependencies:

```
sudo apt install python3-full python3-tk mafft  
pip3 install --break-system-packages setuptools Cython customtkinter biopython pillow matplotlib dill lo
```

Note: `--break-system-packages` may be required on newer Ubuntu releases when installing into the system Python environment.

Step 2. Clone and set up SmarTyper

```
git clone https://github.com/rocpengliu/SmarTyper.git  
cd SmarTyper
```

Step 3. Compile the C++ extension

```
python3 setup.py clean --all && python3 setup.py build_ext --inplace
```

Step 4. Launch SmarTyper

```
python3 smartyper.py
```

Four main functional modules of SmarTyper



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- MACHINE LEARNING
- MICROTYPING
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Welcome to SmarTyper

-- A smart & comprehensive platform for SNP & microtype genotyping, visualization from amplicon sequence data

Genotyping

- Read raw amplicon sequence data (fastq.gz)
- Conduct read quality control
- Demultiplex reads based on locus's primer sequences
- Extract various features
- Predict zygosity & genotype
- Review & correct genotype on interactive genotype plot
- Based on software tool Seq2Type

Open →

Microtyping

- Identify & visualize microtypes (microhaplotype & micropeptide)
- Align sequence & construct phylogenetic tree
- Visualize microtype & sequence alignment
- Visualize phylogenetic tree
- Generate microtype plot
- Based on output genotype file from Genotype module

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Machine Learning

- Train ML models
- Apply models to feature table
- Predict zygosity & genotype for smart genotyping
- Based on an existing genotype table
- This module is optional

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Project Management

- Manage existing projects
- Review & reedit genotype
- Review microtype

Open →

Contact: Dr. Peng Liu
Environment and Climate Change Canada
Email: peng.liu@ec.gc.ca

Citation: Liu, P. et al. SmarTyper: towards smart SNP and microhaplotype genotyping. (2026).
Github: <https://github.com/rocpengliu/SmarTyper>

Step 3 Genotyping module

- This module is designed for automated genotyping by converting raw amplicon sequence data to pre-genotype table based on Seq2Type
- It further provides comprehensive visualization including genotype interactive plot for manual genotype edition and correction

A dark blue rectangular panel with a light blue border. At the top center, the word "Genotyping" is written in a bold, light blue sans-serif font. Below it, a list of seven bullet points is displayed in a smaller, light blue font. At the bottom center, there is a wide, light blue button with the text "Open →" in a dark blue font.

Genotyping

- Read raw amplicon sequence data (fastq.gz)
- Conduct read quality control
- Demultiplex reads based on locus's primer sequences
- Extract various features
- Predict zygosity & genotype
- Review & correct genotype on interactive genotype plot
- Based on software tool Seq2Type

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Step 3.1 Data loading



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Data Loading

Analysis type: ☒ Snp ☐ Sat Select read type

Read type: ☒ PE ☐ SE

Input folder: /home/liup/project/data/prnp/newtest/C Browse and select fastq.gz file folder

Sample file: /home/liup/project/data/prnp/newtest/s: Browse and select sample.txt file

Loc: /home/liup/project/data/prnp/newtest/p: Browse and select loci.txt file

Sex file (optional): ☐ reverse complement

training model (optional): ☐ reverse complement

Output folder: /home/liup/project/data/prnp/newtest/o: Enable this option if your reverse primer seq is not in the same direction of forward primer seq

Id	File Name	Read1	Size1	Read2	Size2	Status
1	20919_S302_L001	20919_S302_L001_R1_001.fastq.gz	3.82 MB	20919_S302_L001_R2_001.fastq.gz	4.17 MB	True
2	21093_S277_L001	21093_S277_L001_R1_001.fastq.gz	2.85 MB	21093_S277_L001_R2_001.fastq.gz	3.02 MB	True
3	21094_S278_L001	21094_S278_L001_R1_001.fastq.gz	3.49 MB	21094_S278_L001_R2_001.fastq.gz	3.70 MB	True
4	21095_S279_L001	21095_S279_L001_R1_001.fastq.gz	3.82 MB	21095_S279_L001_R2_001.fastq.gz	4.17 MB	True
5	21098_S280_L001	21098_S280_L001_R1_001.fastq.gz	2.04 MB	21098_S280_L001_R2_001.fastq.gz	1.84 MB	True
6	21099_S281_L001	21099_S281_L001_R1_001.fastq.gz	2.41 MB	21099_S281_L001_R2_001.fastq.gz	2.59 MB	True
7	21101_S282_L001	21101_S282_L001_R1_001.fastq.gz	2.41 MB	21102_S283_L001_R2_001.fastq.gz	2.59 MB	True
8	21102_S283_L001	21102_S283_L001_R1_001.fastq.gz	3.89 MB	21104_S284_L001_R2_001.fastq.gz	4.31 MB	True
9	21104_S284_L001	21104_S284_L001_R1_001.fastq.gz	4.30 MB	21105_S285_L001_R2_001.fastq.gz	5.36 MB	True
10	21105_S285_L001	21105_S285_L001_R1_001.fastq.gz	4.34 MB	21106_S286_L001_R2_001.fastq.gz	5.36 MB	True
11	21106_S286_L001	21106_S286_L001_R1_001.fastq.gz	3.44 MB	21107_S287_L001_R2_001.fastq.gz	4.05 MB	True
12	21107_S287_L001	21107_S287_L001_R1_001.fastq.gz	2.87 MB	21110_S290_L001_R2_001.fastq.gz	2.91 MB	True
13	21110_S290_L001	21110_S290_L001_R1_001.fastq.gz	3.21 MB	21111_S291_L001_R2_001.fastq.gz	3.71 MB	True
14	21111_S291_L001	21111_S291_L001_R1_001.fastq.gz	3.47 MB	21112_S292_L001_R2_001.fastq.gz	3.79 MB	True
15	21112_S292_L001	21112_S292_L001_R1_001.fastq.gz	3.59 MB	21113_S293_L001_R2_001.fastq.gz	3.58 MB	True
16	21113_S293_L001	21113_S293_L001_R1_001.fastq.gz	4.16 MB	21115_S289_L001_R2_001.fastq.gz	4.48 MB	True
17	21115_S289_L001	21115_S289_L001_R1_001.fastq.gz	4.07 MB	21118_S294_L001_R2_001.fastq.gz	4.82 MB	True
18	21118_S294_L001	21118_S294_L001_R1_001.fastq.gz	3.34 MB	21297_S295_L001_R2_001.fastq.gz	3.34 MB	True
19	21297_S295_L001	21297_S295_L001_R1_001.fastq.gz	3.63 MB	21298_S296_L001_R2_001.fastq.gz	3.87 MB	True
20	21298_S296_L001	21298_S296_L001_R1_001.fastq.gz	3.43 MB	21299_S297_L001_R2_001.fastq.gz	3.58 MB	True
21	21299_S297_L001	21299_S297_L001_R1_001.fastq.gz	3.53 MB	21301_S298_L001_R2_001.fastq.gz	3.82 MB	True
22	21301_S298_L001	21301_S298_L001_R1_001.fastq.gz	2.89 MB	21303_S300_L001_R2_001.fastq.gz	2.89 MB	True
23	21303_S300_L001	21303_S300_L001_R1_001.fastq.gz	2.20 MB	21304_S301_L001_R2_001.fastq.gz	2.39 MB	True
24	21304_S301_L001	21304_S301_L001_R1_001.fastq.gz	3.35 MB	21305_S303_L001_R2_001.fastq.gz	3.84 MB	True
25	21305_S303_L001	21305_S303_L001_R1_001.fastq.gz	2.42 MB	21306_S304_L001_R2_001.fastq.gz	2.76 MB	True
26	21306_S304_L001	21306_S304_L001_R1_001.fastq.gz	2.58 MB	21307_S305_L001_R2_001.fastq.gz	2.86 MB	True
27	21307_S305_L001	21307_S305_L001_R1_001.fastq.gz	4.81 MB	21309_S306_L001_R2_001.fastq.gz	5.13 MB	True
28	21309_S306_L001	21309_S306_L001_R1_001.fastq.gz	4.04 MB	22643_S336_L001_R2_001.fastq.gz	4.31 MB	True
29	22643_S336_L001	22643_S336_L001_R1_001.fastq.gz	3.05 MB	22644_S337_L001_R2_001.fastq.gz	3.03 MB	True
30	22644_S337_L001	22644_S337_L001_R1_001.fastq.gz				

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Browse and select sex.txt file

Browse, create and select output folder and then click 'Confirm' to show your loading data

After loading data checked and confirmed, Click 'Next' to continue

Step 3.2.1 Parameter configuration



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Parameter Setting

General options:

Sequence platform: ☒ Illumina ☐ Nanopore

Min. read length:

Average quality:

Primer mismatches:

Min. reads for locus:

Figure: ☒

Num. thread:

Analysis type: **snp**

smProp1H:

smProp1L:

mmProp1H:

mmProp1L:

mProp2:

Min. reads for allele:

Max. read variants for alignment:

Sex identification options:

Primer mismatches:

Ref. mismatches:

yxRatio:

Min. reads for sex allele:

Min. reads for sex variant:

1. Please specify your sequence platform, illumina is for primer sequence's location is at the beginning of read, otherwise select the nanopore
2. Min. read length: minimum read length for reads quality control
3. Average quality: read quality for quality control
4. Primer mismatches: maximum number of mismatches for demultiplexing reads for each locus' primer sequences
5. Min. reads 4 locus: minimum total number of reads for a locus can be further proceeded
6. Figure: enable figure generation, this could take longer time
7. Num. thread: specify number of thread. If you have limited memory, please decrease number.

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Step 3.2.2 Parameter configuration

Parameter Setting

General options:

- Num. thread: 10
- Min. read length: 30
- Average quality: 20
- Primer mismatches: 4
- Min. reads 4 locus: 30
- Figure: ☒

Analysis type: snp

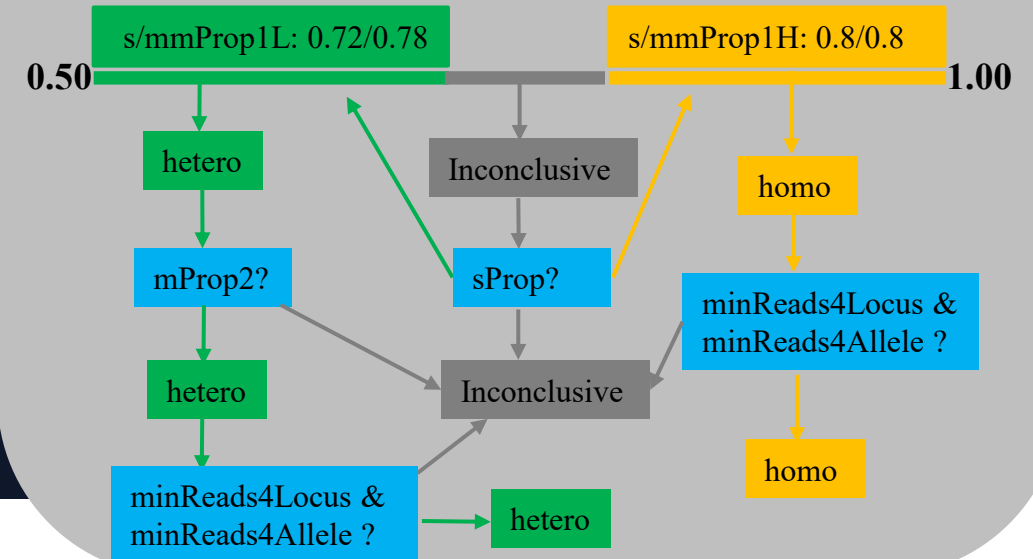
- smProp1H: 0.8
- smProp1L: 0.72
- mmProp1H: 0.8
- mmProp1L: 0.78
- mProp2: 0.7
- Min. reads for allele: 10
- Max. read variants for alignment: 6

sequence variant

	n. reads	name
—————	100	read1/dominant
—————	80	read2/subdominant
—————	10	read3/minor
—————	7	read4
—————	4	read5
—————	2	read6
—————	1	read7
	204	total

$mprop1 = \text{read1} / (\text{read1} + \text{read2}) = 100 / 180 = 0.55$
 $mprop2 = \text{read2} / (\text{read2} + \text{read3}) = 80 / 90 = 0.88$
 $mut = \text{number of mutations between read1 and read2}$
 $sprop = \text{best SNP props (reads with SNP / sum of all read variants)}$

1. smProp1H: threshold for homo prediction when there is a single SNP. Zygosity is predicted as “homo” if $mprop1 \geq smProp1H$
2. smProp1L: threshold for hetero prediction when there is a single SNP. Zygosity is predicted as “hetero” if $mprop1 \leq smProp1L$
3. mmProp1H: threshold for homo prediction when there are multiple SNPs. Zygosity is predicted as “homo” if $mprop1 \geq mmProp1H$
4. mmProp1L: threshold for hetero prediction when there are multiple SNPs. Zygosity is predicted as “hetero” if $mprop1 \geq mmProp1L$
5. mProp2: threshold for hetero prediction. Any hetero predicted by $mprop1$ must also satisfied $mprop2 \geq mProp2$
6. Min. reads for allele: minimum number of reads for a dominant allele (homo) or a subdominant allele (hetero). Otherwise, it is reclassified as “inconclusive”
7. Max. read variants for alignment: maximum number read variants used in alignment table



Step 3.2.3 Parameter configuration



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Parameter Setting

Optional

General options:
Num. thread: 10
Min. read length: 30
Average quality: 20
Primer mismatches: 4
Min. reads 4 locus: 30
Figure: ☒

Analysis type: **snp**
smProp1H: 0.8
smProp1L: 0.72
mmProp1H: 0.8
mmProp1L: 0.78
mProp2: 0.7
Min. reads for allele: 10
Max. read variants for alignment: 6

Sex identification options:
Primer mismatches: 4
Ref. mismatches: 4
yxRatio: 0.0001
Min. reads for sex allele: 10
Min. reads for sex variant: 5

1. Primer mismatches: maximum number of mismatches for both primer sequences
2. Ref. mismatches: maximum number of mismatches for reads against reference sequence
3. Ratio of reads of y/z allele against x/w allele
4. Min. reads for sex allele: minimum number of reads for an allele
5. Min. reads for sex read variant: minimum number of reads for a read variant

After parameters are checked,
Click 'Next' to continue

Step 3.3 Job monitoring

marTyper v1.0

Job Monitoring

processing 2 out of 59 samples with 2 markers (1.69% finished)

Time used: 00h:00m:01s

Estimated remaining time: 00h:00m:28s

Progressing bar

This step may require additional processing time. Depending on your number of samples and loci, it could take several minutes to hours to finish your project. Therefore, please wait until the process completes. Do not close this window!

Job Monitoring

processing 59 out of 59 samples with 2 markers (100%)

Elapsed time: 00h:01m:40s

Success

Seq2Type ran successfully

OK

After finished, click "OK" and then "Next" to proceed

Exit

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Step 3.4 Results review and visualization



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Result Viewing

View results by 'sample' or 'locus'

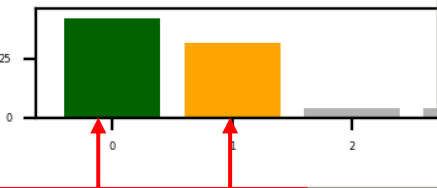
Select individual sample or locus

Sample or Locus: ☒ sample ☐ locus

Geno Combo **Geno Fig & Tab**

Sample Reads All Sample Reads

Sample: 21098_S280_L001, M



Reads proportions of read1 against (read1 + read2) and read2 against (read2 + read3)

Numbers of mutations of top 2 read variants against reference and its subdominant read variant

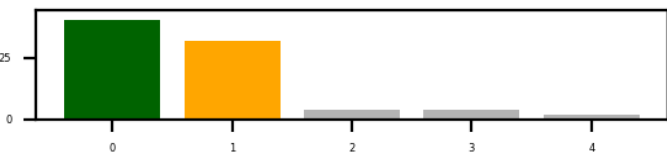
Total reads for this locus

Reads for each read variant

Best read proportion of all the SNVs based on all read variants

	allele	baseChange	readt	read	rprop	mprop	sprop	mut	zygosity	indel	len
PRNP_0			182	42	0.2308	0.5676	0.6296	0 1	heter	N	367
PRNP_1	156(T C)		182	32	0.1758	0.8889	0.6296	1 1	heter	N	367
PRNP_2	77(A G)156(T C)		182	4	0.022	nan	nan	nan	inconclusive	nan	367
PRNP_3	156(T C)248(T C)		182	4	0.022	nan	nan	nan	inconclusive	nan	367
PRNP_4	18(G T)		182	2	0.011	nan	nan	nan	inconclusive	nan	367

Interactive plot for top 5 read variants
Top 2 read variants are highlighted in green and orange



Sequence alignments for each read variant against reference (blue)
Target SNPs are highlighted in green, while new SNPs are highlighted in red. Sequence errors are in orange.

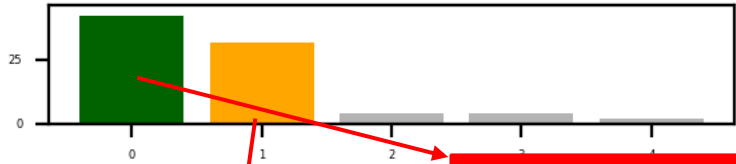
id	sample	allele	baseChange	readt	read	rprop	mprop	sprop	mut	zygosity	indel	len
0	21098_S280_L001	PRNP2_0		178	40	0.2247	0.5556	0.6226	0 1	heter	N	338

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Step 3.4.1 Manual genotype review and correction

Sample: 21098_S280_L001, Marker: PRNP



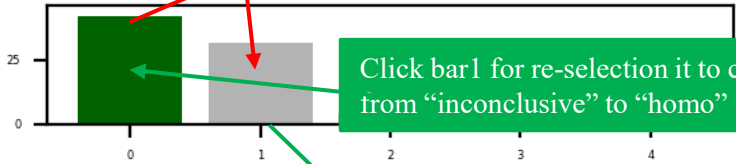
Click bar0 for de-selection both bar0 and bar1 to change zygosity from “heter” to “inconclusive”

Click bar1 for de-selection to change zygosity from “heter” to “homo”

id	sample	allele	baseChange	zygosity	indel	len
0	21098_S280_L001	PRNP_0		heter	N	367
1	21098_S280_L001	PRNP_1	156(T C)	heter	N	367
2	21098_S280_L001	PRNP_2	77(A G)156(T C)	conclusive	nan	367
3	21098_S280_L001	PRNP_3	156(T C)248(T C)	conclusive	nan	367
4	21098_S280_L001	PRNP_4	18(G T)	conclusive	nan	367

Click bar0 for de-selection to change zygosity from “homo” to “inconclusive”

Sample: 21098_S280_L001, Marker: PRNP



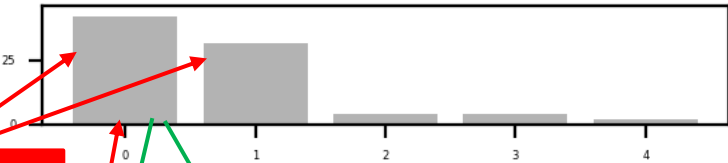
Click bar1 for re-selection it to change zygosity from “inconclusive” to “homo”

id	sample	allele	baseChange
0	21098_S280_L001	PRNP_0	
1	21098_S280_L001	PRNP_1	156(T C)
2	21098_S280_L001	PRNP_2	77(A G)156(T C)
3	21098_S280_L001	PRNP_3	156(T C)248(T C)
4	21098_S280_L001	PRNP_4	18(G T)

Click bar1 for re-selection it to change zygosity from “homo” to “heter”

GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTAT
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC

Sample: 21098_S280_L001, Marker: PRNP

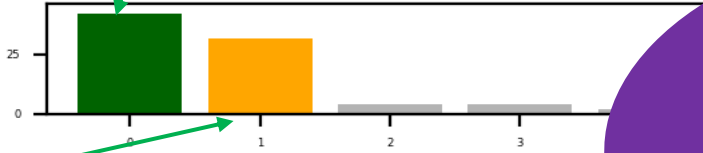


Click bar0 for re-selection both bar0 and bar1 to change zygosity from “inconclusive” to “heter”

id	sample	allele	baseChange	zygosity	indel	len
0	21098_S280_L001	PRNP_0		heter	N	367
1	21098_S280_L001	PRNP_1	156(T C)	heter	N	367
2	21098_S280_L001	PRNP_2	77(A G)156(T C)	conclusive	nan	367
3	21098_S280_L001	PRNP_3	156(T C)248(T C)	conclusive	nan	367
4	21098_S280_L001	PRNP_4	18(G T)	conclusive	nan	367

Click bar0 for re-selection both bar0 and bar1 to change zygosity from “inconclusive” to “heter”

Sample: 21098_S280_L001, Marker: PRNP

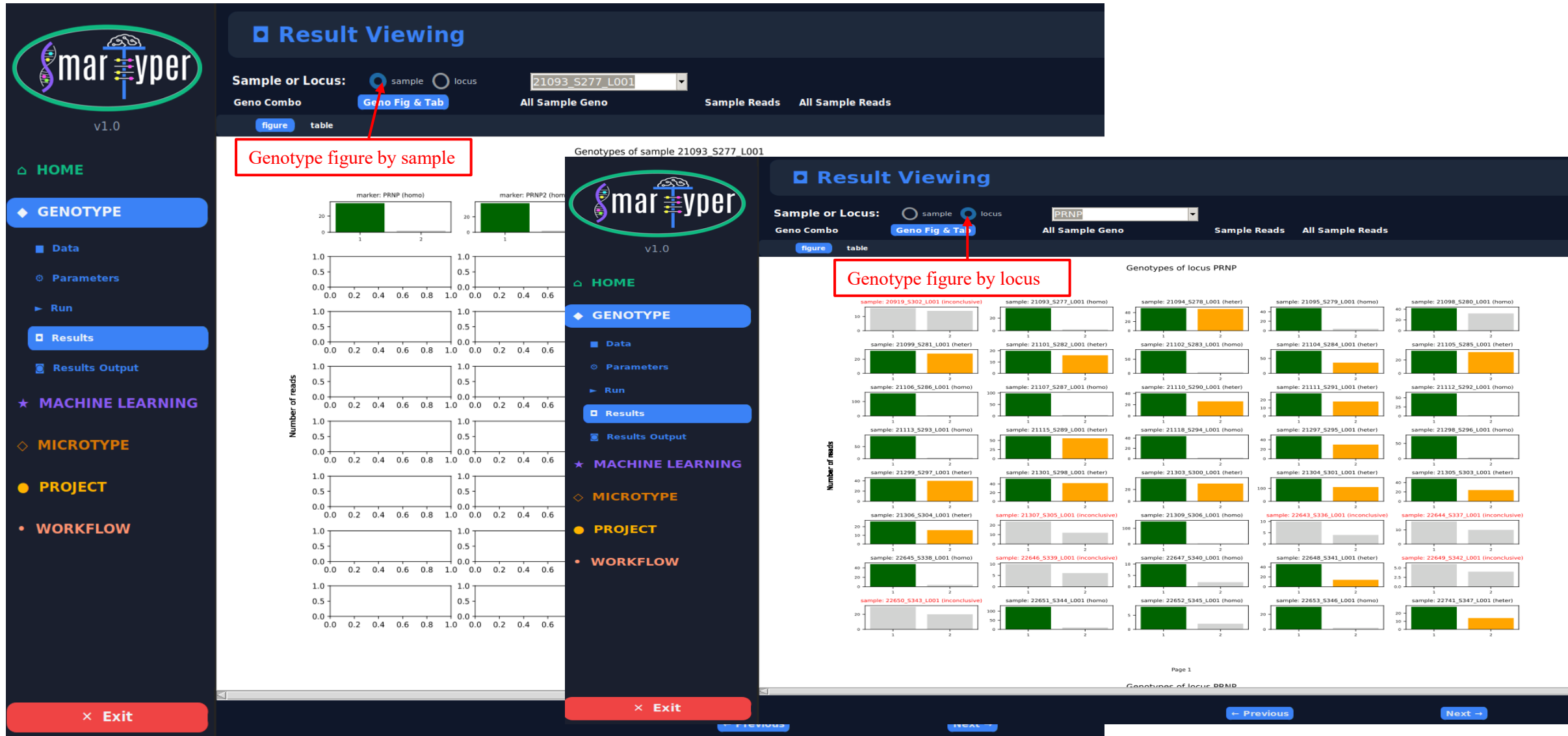


id	sample	allele	baseChange
0	21098_S280_L001	PRNP_0	
1	21098_S280_L001	PRNP_1	156(T C)
2	21098_S280_L001	PRNP_2	77(A G)156(T C)
3	21098_S280_L001	PRNP_3	156(T C)248(T C)
4	21098_S280_L001	PRNP_4	18(G T)

GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC
GGGCCTCGGTGGCTACATCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATC

Carefully review every locus and sample to confirm or manually correct genotype calls

Step 3.4.2.1 Genotype figure (sample or locus)



Step 3.4.2.2 Genotype table (sample or locus)

Result Viewing

Sample or Locus: ☒ sample ☐ locus

21093_S277_L001

Geno Combo

Geno Fig & Tab

All Sample Geno

Sample Reads

All Sample Reads

figure **table**

Id	sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity	baseChange
0	21093_S277_L001	PRNP	1									
1	21093_S277_L001	PRNP	2									
2	21093_S277_L001	PRNP2	1									
3	21093_S277_L001	PRNP2	2									

Genotype table by sample

Result Viewing

Sample or Locus: ☐ sample ☒ locus

PRNP

Geno Combo

Geno Fig & Tab

All Sample Geno

Sample Reads

All Sample Reads

figure **table**

Id	sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity
0	20919_S302_L001	PRNP	1	66	16	0.2424	0.5333	0.7368	1 2	N	inconclusive
1	20919_S302_L001	PRNP	2	66	14	0.2121	0.7778	0.7368	1 2	N	inconclusive
2	21093_S277_L001	PRNP	1	66	36	0.5455	0.9474	0.9375	1 0	N	homo
3	21093_S277_L001	PRNP	2	66	2	0.0303	0.5	0.9375	0 0	N	homo
4	21094_S278_L001	PRNP	1	230	50	0.2174	0.5102	0.5294	1 1	N	heter
5	21094_S278_L001	PRNP	2	230	48	0.2087	0.96	0.5294	0 1	N	heter
6	21095_S279_L001	PRNP	1	192	48	0.25	0.9231	0.9718	0 0	N	homo
7	21095_S279_L001	PRNP	2	192	4	0.0208	0.5	0.9718	0 0	N	homo
8	21098_S280_L001	PRNP	1	182	42	0.2308	0.5676	0.6296	0 1	N	homo
9	21098_S280_L001	PRNP	2	182	32	0.1758	0.8889	0.6296	1 1	N	homo
10	21099_S281_L001	PRNP	1	160	32	0.2	0.5333	0.5686	1 2	N	heter
11	21099_S281_L001	PRNP	2	160	28	0.175	0.875	0.5686	1 2	N	heter
12	21101_S282_L001	PRNP	1	72	20	0.2778	0.5556	0.5	0 1	N	heter
13	21101_S282_L001	PRNP	2	72	16	0.2222	0.8889	0.5	1 1	N	heter
14	21102_S283_L001	PRNP	1	192	80	0.4167	0.9524	0.9818	1 0	N	homo
15	21102_S283_L001	PRNP	2	192	4	0.0208	0.5	0.9818	0 0	N	homo
16	21104_S284_L001	PRNP	1	290	76	0.2621	0.6786	0.6067	1 1	N	heter
17	21104_S284_L001	PRNP	2	290	36	0.1241	0.9	0.6067	0 1	N	heter
18	21105_S285_L001	PRNP	1	214	34	0.1589	0.5152	0.5429	0 2	N	heter
19	21105_S285_L001	PRNP	2	214	32	0.1495	0.8	0.5429	2 2	N	heter
20	21106_S286_L001	PRNP	1	336	156	0.4643	0.975	0.9775	1 0	N	homo
21	21106_S286_L001	PRNP	2	336	4	0.0119	0.5	0.9775	0 0	N	homo
22	21107_S287_L001	PRNP	1	262	98	0.374	0.9608	0.9877	0 0	N	homo
23	21107_S287_L001	PRNP	2	262	4	0.0153	0.5	0.9877	0 0	N	homo
24	21110_S290_L001	PRNP	1	158	40	0.2532	0.6061	0.5208	0 1	N	heter
25	21110_S290_L001	PRNP	2	158	26	0.1646	0.9286	0.5208	1 1	N	heter
26	21111_S291_L001	PRNP	1	132	28	0.2121	0.6087	0.5814	0 1	N	heter
27	21111_S291_L001	PRNP	2	132	18	0.1364	0.8182	0.5814	1 1	N	heter
28	21112_S292_L001	PRNP	1	124	62	0.5	0.9688	0.9375	1 0	N	homo
29	21112_S292_L001	PRNP	2	124	2	0.0161	0.5	0.9375	0 0	N	homo
30	21113_S293_L001	PRNP	1	206	94	0.4563	0.9592	0.9636	1 0	N	homo
31	21113_S293_L001	PRNP	2	206	4	0.0194	0.5	0.9636	0 0	N	homo
32	21115_S289_L001	PRNP	1	220	62	0.2818	0.5254	0.5769	1 1	N	heter
33	21115_S289_L001	PRNP	2	220	56	0.2545	0.9333	0.5769	0 1	N	heter

Genotype table by locus

Exit

Exit

Previous

Next

Step 3.4.3 Genotype tables (all samples or loci)


v1.0

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Result Viewing

Sample or Locus: ☐ sample ☒ locus

Geno Combo Geno Fig & Tab **All Sample Geno** Sample Reads All Sample Reads

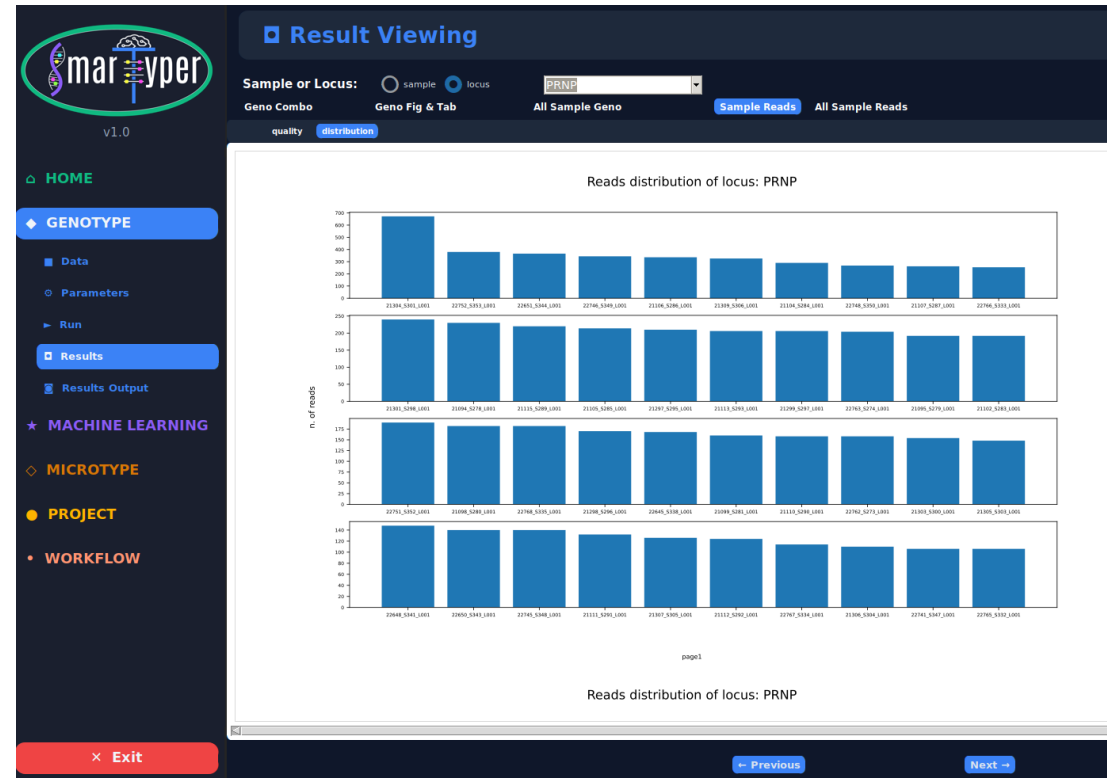
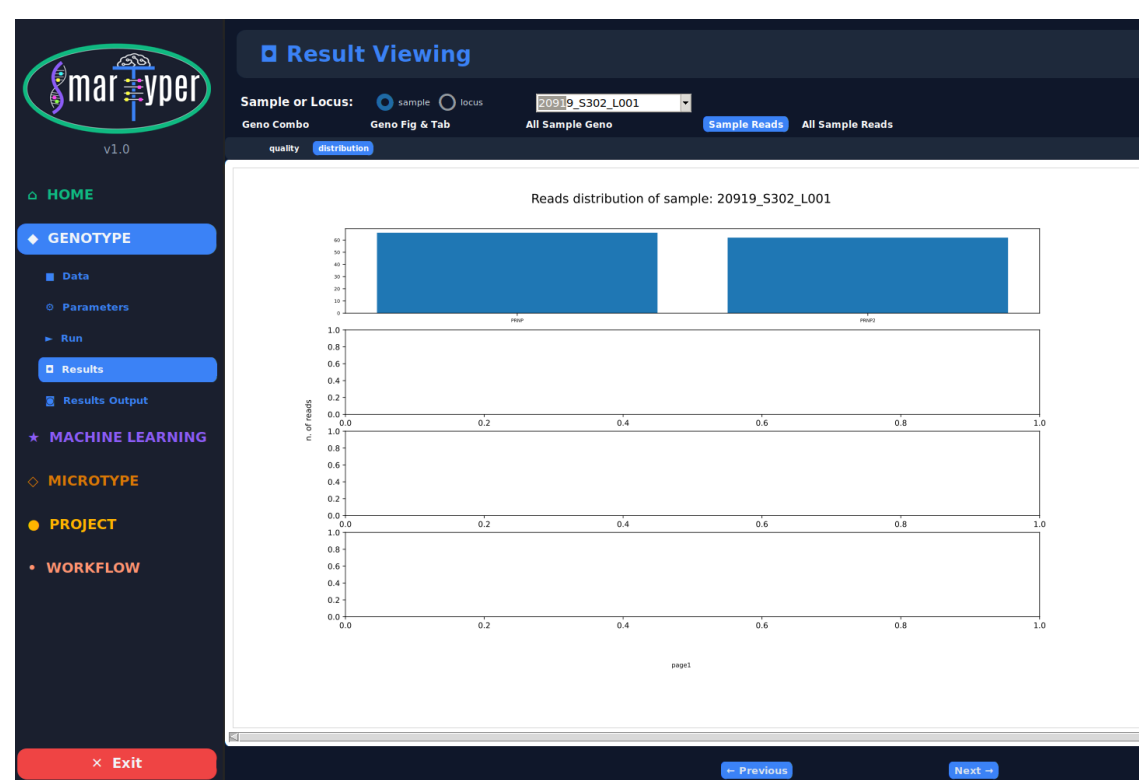
Id	sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity	baseChange
0	20919_S302_L001	PRNP	1	66	16	0.2424	0.5333	0.7368	1 2	N	inconclusive	347(C T)
1	20919_S302_L001	PRNP	2	66	14	0.2121	0.7778	0.7368	1 2	N	inconclusive	60(C T)
2	20919_S302_L001	PRNP2	1	62	16	0.2581	0.5333	0.7059	1 2	N	inconclusive	334(C T)
3	20919_S302_L001	PRNP2	2	62	14	0.2258	0.7778	0.7059	1 2	N	inconclusive	47(C T)
4	21093_S277_L001	PRNP	1	66	36	0.5455	0.9474	0.9375	1 0	N	homo	60(C T)
5	21093_S277_L001	PRNP	2	66	2	0.0303	0.5	0.9375	0 0	N	homo	5(T C)60(C T)
6	21093_S277_L001	PRNP2	1	66	38	0.5758	0.95	0.9333	1 0	N	homo	47(C T)
7	21093_S277_L001	PRNP2	2	66	2	0.0303	0.5	0.9333	0 0	N	homo	47(C T)141(T C)
8	21094_S278_L001	PRNP	1	230	50	0.2174	0.5102	0.5294	1 1	N	heter	7(G A)
9	21094_S278_L001	PRNP	2	230	48	0.2087	0.96	0.5294	0 1	N	heter	
10	21094_S278_L001	PRNP2	1	226	98	0.4336	0.9608	0.9836	0 0	N	homo	
11	21094_S278_L001	PRNP2	2	226	4	0.0177	0.5	0.9836	0 0	N	homo	108(A G)
12	21095_S279_L001	PRNP	1	192	48	0.25	0.9231	0.9718	0 0	N	homo	
13	21095_S279_L001	PRNP	2	192	4	0.0208	0.5	0.9718	0 0	N	homo	279(C T)316(G A)
14	21095_S279_L001	PRNP2	1	186	54	0.2903	0.931	0.9846	0 0	N	homo	
15	21095_S279_L001	PRNP2	2	186	4	0.0215	0.5	0.9846	0 0	N	homo	266(C T)303(G A)
16	21098_S280_L001	PRNP	1	182	42	0.2308	0.5676	0.6296	0 1	N	homo	
17	21098_S280_L001	PRNP	2	182	32	0.1758	0.8889	0.6296	1 1	N	homo	156(T C)
18	21098_S280_L001	PRNP2	1	178	40	0.2247	0.5556	0.6226	0 1	N	heter	
19	21098_S280_L001	PRNP2	2	178	32	0.1798	0.8889	0.6226	1 1	N	heter	143(T C)
20	21099_S281_L001	PRNP	1	160	32	0.2	0.5333	0.5686	1 2	N	heter	60(C T)
21	21099_S281_L001	PRNP	2	160	28	0.175	0.875	0.5686	1 2	N	heter	156(T C)
22	21099_S281_L001	PRNP2	1	158	34	0.2152	0.5484	0.551	1 2	N	heter	47(C T)
23	21099_S281_L001	PRNP2	2	158	28	0.1772	0.875	0.551	1 2	N	heter	143(T C)
24	21101_S282_L001	PRNP	1	72	20	0.2778	0.5556	0.5	0 1	N	heter	
25	21101_S282_L001	PRNP	2	72	16	0.2222	0.8889	0.5	1 1	N	heter	347(C T)
26	21101_S282_L001	PRNP2	1	72	22	0.3056	0.5789	0.5263	0 1	N	heter	
27	21101_S282_L001	PRNP2	2	72	16	0.2222	0.8889	0.5263	1 1	N	heter	334(C T)
28	21102_S283_L001	PRNP	1	192	80	0.4167	0.9524	0.9818	1 0	N	homo	7(G A)
29	21102_S283_L001	PRNP	2	192	4	0.0208	0.5	0.9818	0 0	N	homo	7(G A)14(A G)
30	21102_S283_L001	PRNP2	1	190	84	0.4421	0.9545	0.9808	0 0	N	homo	
31	21102_S283_L001	PRNP2	2	190	4	0.0211	0.5	0.9808	0 0	N	homo	1(A G)
32	21104_S284_L001	PRNP	1	290	76	0.2621	0.6786	0.6067	1 1	N	heter	347(C T)
33	21104_S284_L001	PRNP	2	290	36	0.1241	0.9	0.6067	0 1	N	heter	
34	21104_S284_L001	PRNP2	1	286	78	0.2727	0.65	0.6145	1 1	N	heter	334(C T)

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Step 3.4.4.1 Reads quality (sample)



Step 3.4.4.2 Read distribution (sample or locus)

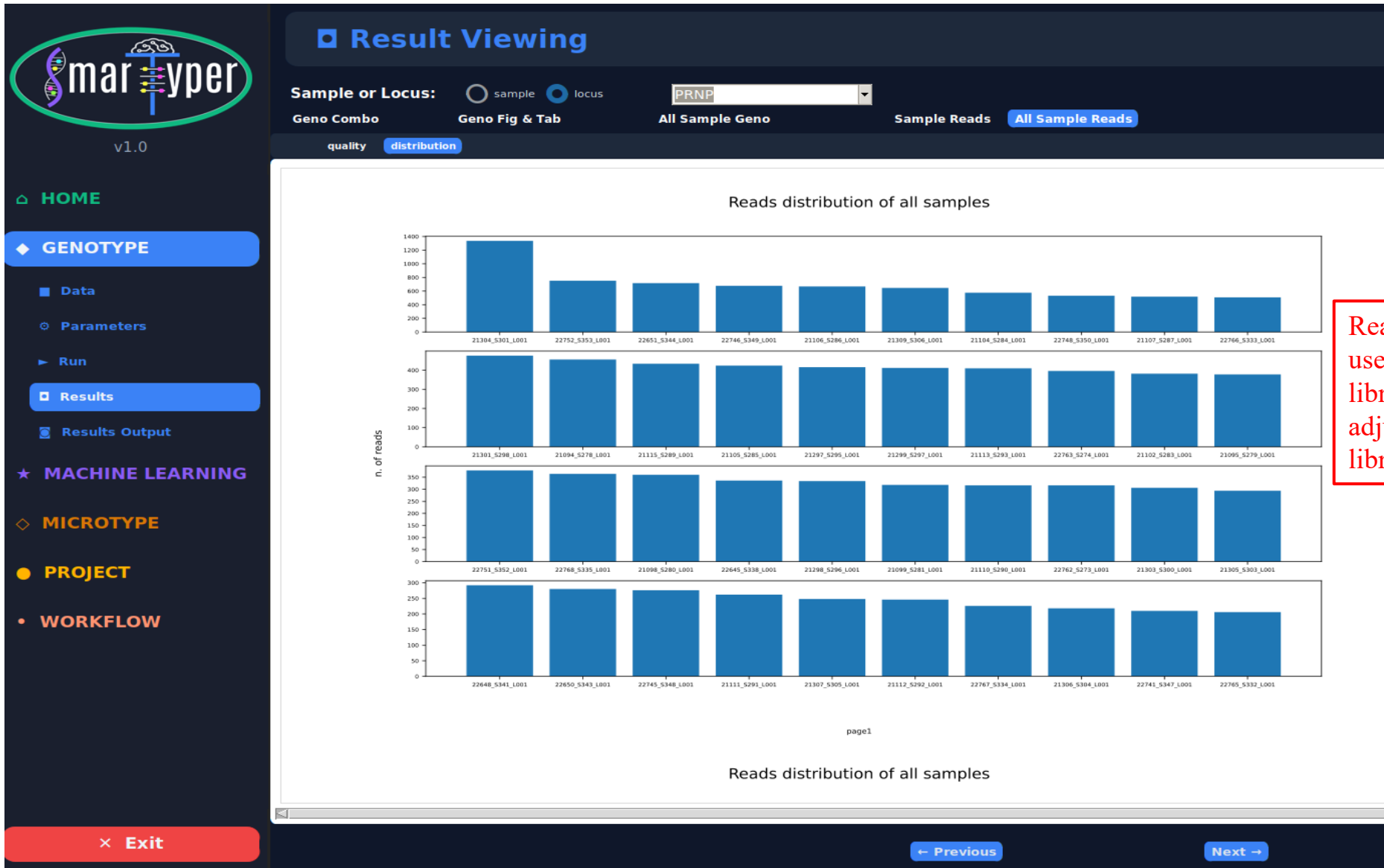


Read distribution among loci for one sample can help optimize your multiplex PCR, read distribution among samples for one locus could help assess the primer's efficiency

Step 3.4.3.1 Read quality (all samples)



Step 3.4.3.2 Read distribution across all samples



Read distribution among samples are used to optimize quality of your library pooling, you may need to adjust among of PCR product for your library mixing.

Step 3.4.4.1 File outputs

The screenshot displays the marTyper v1.0 software interface. On the left is a dark sidebar with a logo at the top and a menu of options: HOME, GENOTYPE (selected), MACHINE LEARNING, MICROTYPE, PROJECT, and WORKFLOW. The GENOTYPE menu is expanded, showing sub-options: Data, Parameters, Run, Results, and Results Output (highlighted). At the bottom of the sidebar is a red 'Exit' button.

The main window has a dark header with a 'Result Output' title and a message: 'This is to output all tables and figures to local storage! Please be patient, do not close the program.' Below this, a log window shows the following text:

```
Running final analysis and generating summary...
[15:36:25]: This process is very memory intensive, it may take a while to finish. Please be patient and do not close the program.

[15:36:25]: -----start step 1-----
[15:36:25]: starting to dump genotyping project!
[15:36:25]: This is slow, please be patient!
[15:36:25]: finished to dump genotyping project!
[15:36:25]: -----end step 1-----

[15:36:25]: -----
[15:36:25]: starting to output
[15:36:25]: starting to process
```

Below the log window is a smaller version of the marTyper v1.0 interface, showing the same menu and a 'Results Output' button. At the bottom of the main window are 'Previous' and 'Microtype' buttons.

This function is used to output all results including tables, figures

Step 3.4.4.2 Key output files

1. All_sample_final_haplotype.txt, used for microtype genotyping

sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity	baseChange	mh_seq	id
20919_S302_L001	PRNP	1	66	16	0.2424	0.5333	0.7368	1 2	N	inconclusive	347(C T)	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCA	
20919_S302_L001	PRNP	2	66	14	0.2121	0.7778	0.7368	1 2	N	inconclusive	60(C T)	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGG	
20919_S302_L001	PRNP2	1	62	16	0.2581	0.5333	0.7059	1 2	N	inconclusive	334(C T)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
20919_S302_L001	PRNP2	2	62	14	0.2258	0.7778	0.7059	1 2	N	inconclusive	47(C T)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21093_S277_L001	PRNP	1	66	36	0.5455	0.9474	0.9375	1 0	N	homo	60(C T)	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21093_S277_L001	PRNP	2	66	2	0.0303	0.5	0.9375	0 0	N	homo	5(T C)60(C T)	GGGCCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGG	
21093_S277_L001	PRNP2	1	66	38	0.5758	0.95	0.9333	1 0	N	homo	47(C T)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21093_S277_L001	PRNP2	2	66	2	0.0303	0.5	0.9333	0 0	N	homo	47(C T)141(T C)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21094_S278_L001	PRNP	1	230	50	0.2174	0.5102	0.5294	1 1	N	heter	7(G A)	GGGCCTCAGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21094_S278_L001	PRNP	2	230	48	0.2087	0.96	0.5294	0 1	N	heter		GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21094_S278_L001	PRNP2	1	226	98	0.4336	0.9608	0.9836	0 0	N	homo		TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21094_S278_L001	PRNP2	2	226	4	0.0177	0.5	0.9836	0 0	N	homo	108(A G)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21095_S279_L001	PRNP	1	192	48	0.25	0.9231	0.9718	0 0	N	homo		GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21095_S279_L001	PRNP	2	192	4	0.0208	0.5	0.9718	0 0	N	homo	279(C T)316(G A)	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCA	
21095_S279_L001	PRNP2	1	186	54	0.2903	0.931	0.9846	0 0	N	homo		TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21095_S279_L001	PRNP2	2	186	4	0.0215	0.5	0.9846	0 0	N	homo	266(C T)303(G A)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCT	

2. All_sample_final_ml.txt, used for machine learning training

sample	locus	readt	read1	read2	read3	rprop1	rprop2	rprop3	mprop1	mprop2	sprop	mut	indel	zygosity
20919_S302_L001	PRNP	66	16	14	4	0.2424	0.2121	0.0606	0.5333	0.7778	0.7368	2	0	0
22652_S345_L001	PRNP	22	8	2	2	0.3636	0.0909	0.0909	0.8	0.5	0.875	0	0	1
22651_S344_L001	PRNP	366	124	10	4	0.3388	0.0273	0.0109	0.9254	0.7143	0.9469	0	0	1
22650_S343_L001	PRNP	140	30	20	6	0.2143	0.1429	0.0429	0.6	0.7692	0.5111	1	0	0
22649_S342_L001	PRNP	26	6	4	2	0.2308	0.1538	0.0769	0.6	0.6667	0.5	1	0	0
22648_S341_L001	PRNP	148	46	14	2	0.3108	0.0946	0.0135	0.7667	0.875	0.5435	1	0	2
22647_S340_L001	PRNP	28	10	2	2	0.3571	0.0714	0.0714	0.8333	0.5	0.9	0	0	1
22646_S339_L001	PRNP	70	10	6	2	0.1429	0.0857	0.0286	0.625	0.75	0.5172	1	0	0
22653_S346_L001	PRNP	72	30	2	2	0.4167	0.0278	0.0278	0.9375	0.5	0.9545	0	0	1
22645_S338_L001	PRNP	168	48	4	2	0.2857	0.0238	0.0119	0.9231	0.6667	0.9833	0	0	1
22643_S336_L001	PRNP	42	10	4	2	0.2381	0.0952	0.0476	0.7143	0.6667	0.625	2	0	0
21309_S306_L001	PRNP	326	144	4	4	0.4417	0.0123	0.0123	0.973	0.5	0.9889	0	0	1
21307_S305_L001	PRNP	126	24	12	4	0.1905	0.0952	0.0317	0.6667	0.75	0.8636	1	0	0
21306_S304_L001	PRNP	110	26	16	4	0.2364	0.1455	0.0364	0.619	0.8	0.5143	1	0	2
21305_S303_L001	PRNP	148	48	24	4	0.3243	0.1622	0.027	0.6667	0.8571	0.5128	1	0	2
21304_S301_L001	PRNP	672	174	110	4	0.2589	0.1637	0.006	0.6127	0.9649	0.5363	1	0	2
21303_S300_L001	PRNP	154	38	30	2	0.2468	0.1948	0.013	0.5588	0.9375	0.5333	1	0	2
22768_S335_L001	PRNP	182	44	24	4	0.2418	0.1319	0.022	0.6471	0.8571	0.5263	1	0	2
21301_S298_L001	PRNP	240	52	42	4	0.2167	0.175	0.0167	0.5532	0.913	0.507	1	0	2
22741_S347_L001	PRNP	106	28	14	2	0.2642	0.1321	0.0189	0.6667	0.875	0.6471	1	0	2
22746_S349_L001	PRNP	344	78	66	4	0.2267	0.1919	0.0116	0.5417	0.9429	0.5152	1	0	2

3. smartyper_log.txt, useful for debugging Seq2Type

```
[16:08:06]: Starting Seq2Type for genotyping!
[16:08:06]: -----start index: 1-----
[16:08:06]: Start to process sample: 20919_S302_L001, 1 out of 59 samples
[16:08:06]: Running Seq2Type for sample: 20919_S302_L001, this is slow and please be patient!
[16:08:06]: Reading locus file: /home/liup/project/data/prnp/newtest/prnp_overlapped.txt
[16:08:07]: Read loci of: 2
[16:08:07]: start to load data
[16:08:08]: all reads loaded, start to monitor thread status
[16:08:08]: reading sample 20919_S302_L001 outputs
[16:08:08]: reading sample 20919_S302_L001 genotype done!
[16:08:08]: reading sample 20919_S302_L001 reads done!
[16:08:08]: reading sample 20919_S302_L001 json file done!
[16:08:08]: plotting sample 20919_S302_L001 reads quality done!
[16:08:08]: Finish the processing sample: 20919_S302_L001, 1 out of 59 samples
[16:08:08]: -----end index: 1-----
```

Step 4 Microtype genotyping

This module is designed for microtype including microhaplotype (DNA level) and micropeptide (protein level) genotyping and visualization

Microtyping

- Identify & visualize microtypes (microhaplotype & micropeptide)
- Align sequence & construct phylogenetic tree
- Visualize microtype & sequence alignment
- Visualize phylogenetic tree
- Generate microtype plot
- Based on output genotype file from Genotype module

Open →

Step 4.1 Microtype data loading



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◇ Microtype Loading

Num. thread:

Loci file: ☐ reverse complement

Microtype input file:

Pre microhap file (optional):

Output folder:

[15:49:12]: -----start step 7-----

[15:49:12]: starting to generate microhaps figs for each sample!

[15:49:17]: Finished processing 10 out of 59 samples for microhap plotting...

[15:49:18]: Finished processing 20 out of 59 samples for microhap plotting...

[15:49:19]: Finished processing 30 out of 59 samples for microhap plotting...

[15:49:21]: Finished processing 40 out of 59 samples for microhap plotting...

[15:49:22]: Finished processing 50 out of 59 samples for microhap plotting...

[15:49:23]: finished to generate microtypes figs for each sample!

[15:49:23]: -----end step 7-----

[15:49:23]: -----start step 8-----

[15:49:23]: Starting to dump session for project: microtype

[15:49:24]: Session successfully saved to: /home/liup/project/data/prnp/newtest/mh_out1/project_microtype_session.dill

[15:49:24]: -----end step 8-----

[15:49:24]: Microhap processing completed successfully. Please click 'Next' to proceed.

← Previous Next →

Same loci file as used for "Genotyping module"

All_sample_final_haplotype.txt file from output of "Genotype module"

All_mt_lookup_table.txt file from output of previous "Microtype module" (optional)

Specify or create a new output folder

Success

✓

Success

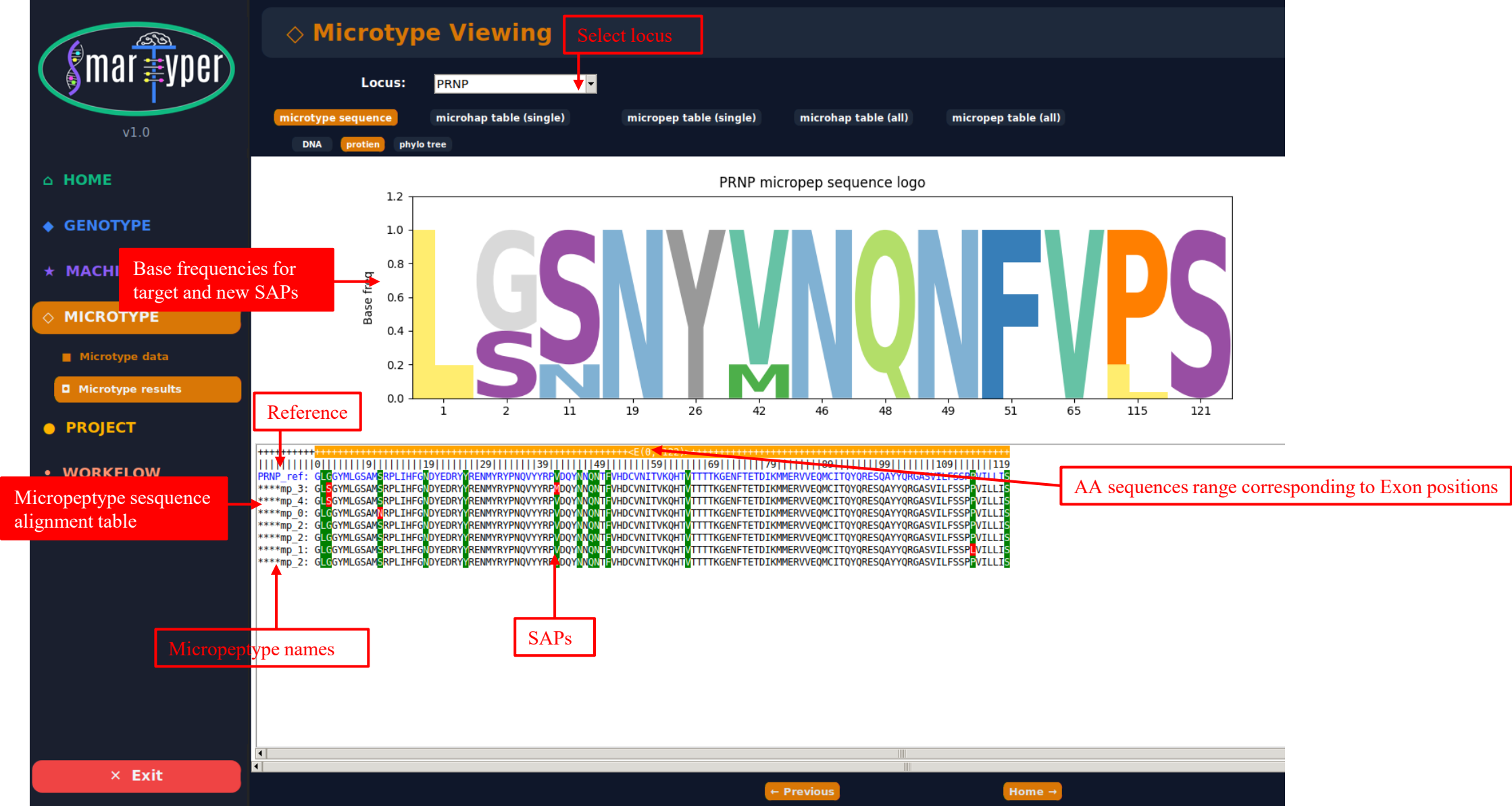
Microhap processing completed successfully. Please click 'Next' to proceed.

OK

Step 4.2.1.1 Microtype visualization (microhaplotypes)



Step 4.2.1.2 Microtype visualization (micropeptides)



Step 4.2.1.3 Phylogenetic trees



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◇ Microtype Viewing

Locus: PRNP

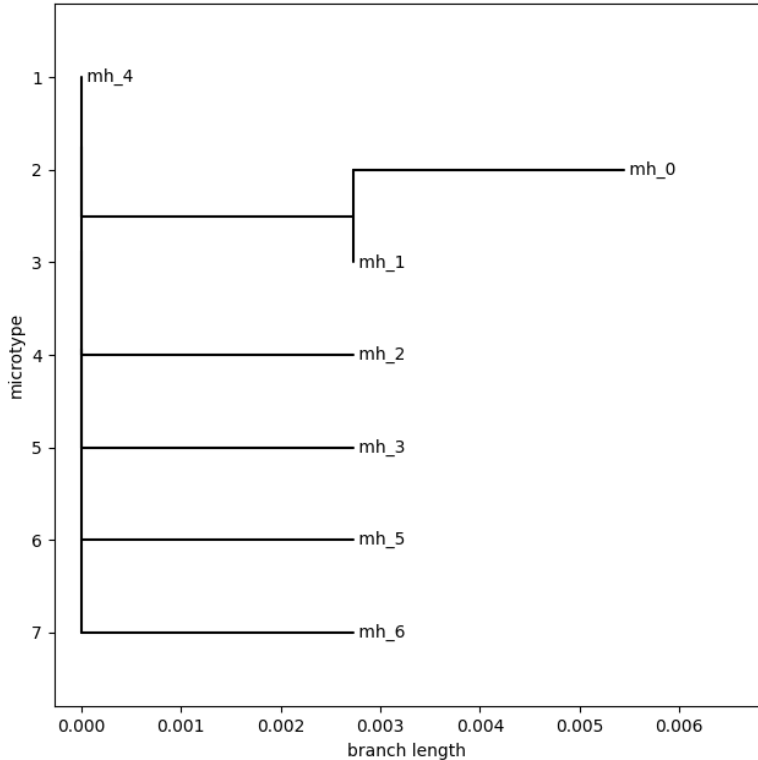
microtype sequence microhap table (single) micropep table (single) microhap table (all) micropep table (all)

DNA protien **phylo tree**

Phylogenetic Trees PRNP

Microhaplotype tree

DNA tree

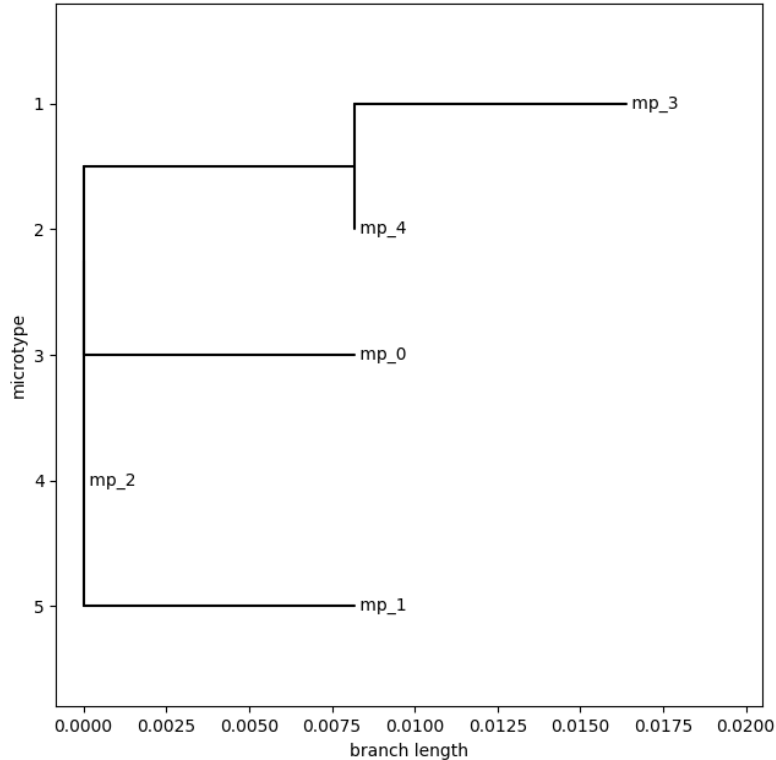


microtype

branch length

Micropeptide tree

AA tree



microtype

branch length

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Step 4.2.2.1 Comprehensive MH table (locus)



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Microtype Viewing

Locus: PRNP

microtype sequence microhap table (single) micropep table (single) microhap table (all) micropep table (all)

comprehensive simple figure

id	sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity	baseChange
0	20919_S302_L001	PRNP	mh_5	66	16	0.2424	0.5333	0.7368	1 2	N	inconclusive	347(C T)
1	20919_S302_L001	PRNP	mh_6	66	14	0.2121	0.7778	0.7368	1 2	N	inconclusive	60(C T)
2	21093_S277_L001	PRNP	mh_6	66	36	0.5455	0.9474	0.9375	1 0	N	homo	60(C T)
3	21093_S277_L001	PRNP	-9	66	2	0.0303	0.5	0.9375	0 0	N	homo	5(T C)60(C T)
4	21094_S278_L001	PRNP	mh_1	230	50	0.2174	0.5102	0.5294	1 1	N	heter	7(G A)
5	21094_S278_L001	PRNP	mh_4	230	48	0.2087	0.96	0.5294	0 1	N	heter	nan
6	21095_S279_L001	PRNP	mh_4	192	48	0.25	0.9231	0.9718	0 0	N	homo	nan
7	21095_S279_L001	PRNP	-9	192	4	0.0208	0.5	0.9718	0 0	N	homo	279(C T)316(G A)
8	21098_S280_L001	PRNP	mh_4	182	42	0.2308	0.5676	0.6296	0 1	N	heter	nan
9	21098_S280_L001	PRNP	mh_3	182	32	0.1758	0.8889	0.6296	1 1	N	heter	156(T C)
10	21099_S281_L001	PRNP	mh_6	160	32	0.2	0.5333	0.5686	1 2	N	heter	60(C T)
11	21099_S281_L001	PRNP	mh_3	160	28	0.175	0.875	0.5686	1 2	N	heter	156(T C)
12	21101_S282_L001	PRNP	mh_4	72	20	0.2778	0.5556	0.5	0 1	N	heter	nan
13	21101_S282_L001	PRNP	mh_5	72	16	0.2222	0.8889	0.5	1 1	N	heter	347(C T)
14	21102_S283_L001	PRNP	mh_1	192	80	0.4167	0.9524	0.9818	1 0	N	homo	7(G A)
15	21102_S283_L001	PRNP	-9	192	4	0.0208	0.5	0.9818	0 0	N	homo	7(G A)14(A G)
16	21104_S284_L001	PRNP	mh_5	290	76	0.2621	0.6786	0.6067	1 1	N	heter	347(C T)
17	21104_S284_L001	PRNP	mh_4	290	36	0.1241	0.9	0.6067	0 1	N	heter	nan
18	21105_S285_L001	PRNP	mh_4	214	34	0.1589	0.5152	0.5429	0 2	N	heter	nan
19	21105_S285_L001	PRNP	mh_0	214	32	0.1495	0.8	0.5429	2 2	N	heter	7(G A)127(G A)
20	21106_S286_L001	PRNP	mh_6	336	156	0.4643	0.975	0.9775	1 0	N	homo	60(C T)
21	21106_S286_L001	PRNP	-9	336	4	0.0119	0.5	0.9775	0 0	N	homo	60(C T)293(A G)
22	21107_S287_L001	PRNP	mh_4	262	98	0.374	0.9608	0.9877	0 0	N	homo	nan
23	21107_S287_L001	PRNP	-9	262	4	0.0153	0.5	0.9877	0 0	N	homo	287(A G)
24	21110_S290_L001	PRNP	mh_4	158	40	0.2532	0.6061	0.5208	0 1	N	heter	nan
25	21110_S290_L001	PRNP	mh_3	158	26	0.1646	0.9286	0.5208	1 1	N	heter	156(T C)
26	21111_S291_L001	PRNP	mh_4	132	28	0.2121	0.6087	0.5814	0 1	N	heter	nan
27	21111_S291_L001	PRNP	mh_1	132	18	0.1364	0.8182	0.5814	1 1	N	heter	7(G A)
28	21112_S292_L001	PRNP	mh_6	124	62	0.5	0.9688	0.9375	1 0	N	homo	60(C T)
29	21112_S292_L001	PRNP	-9	124	2	0.0161	0.5	0.9375	0 0	N	homo	48(A G)60(C T)
30	21113_S293_L001	PRNP	mh_6	206	94	0.4563	0.9592	0.9636	1 0	N	homo	60(C T)
31	21113_S293_L001	PRNP	-9	206	4	0.0194	0.5	0.9636	0 0	N	homo	60(C T)291(A G)
32	21115_S289_L001	PRNP	mh_1	220	62	0.2818	0.5254	0.5769	1 1	N	heter	7(G A)
33	21115_S289_L001	PRNP	mh_4	220	56	0.2545	0.9333	0.5769	0 1	N	heter	nan

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Step 4.2.2.2 Simple MH table (locus)


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Microtype Viewing

Locus: PRNP

microtype sequence **microhap table (single)** micropep table (single) microhap table (all) micropep table (all)

comprehensive **simple** figure

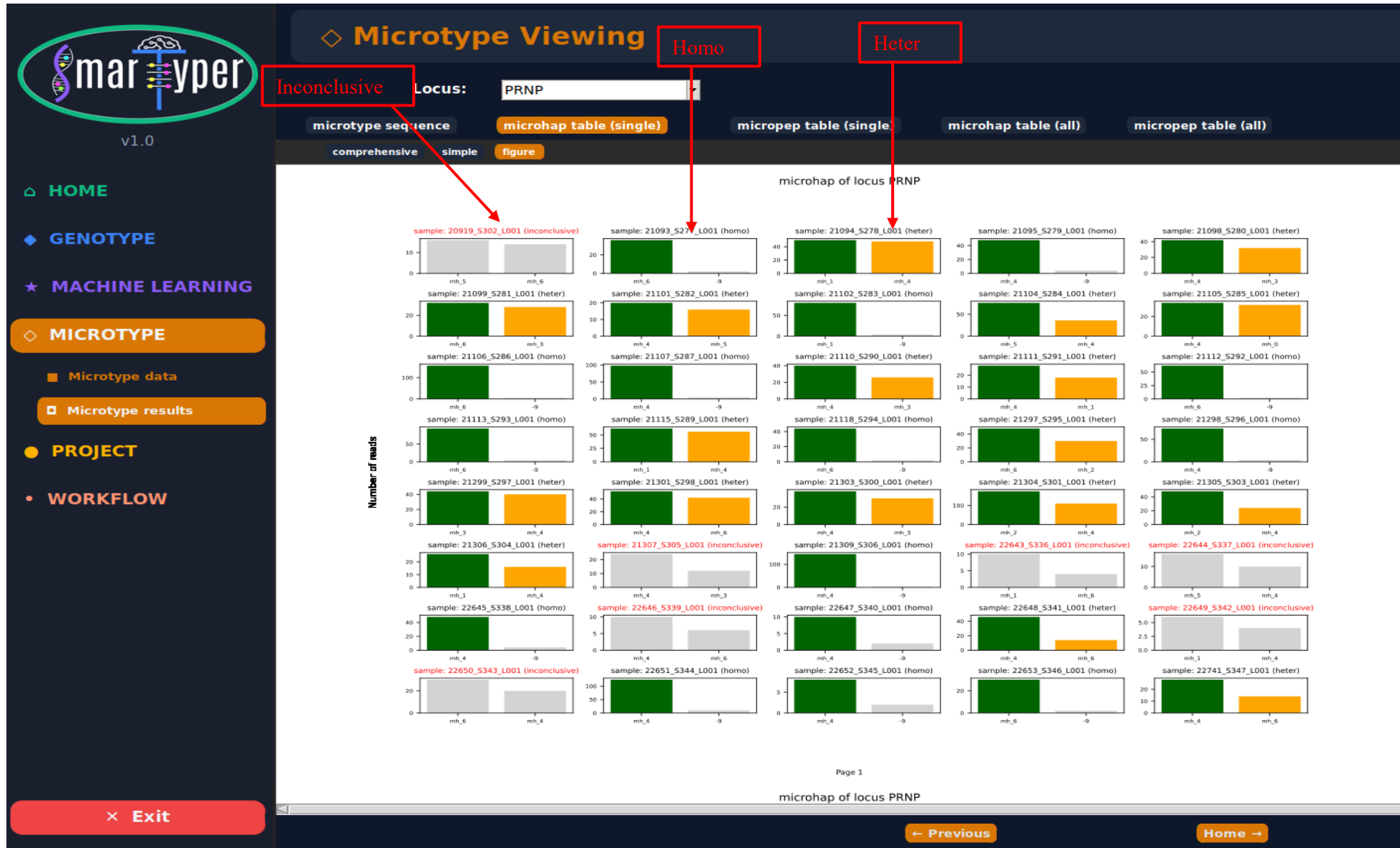
id	sample	locus	allele1	allele2
0	20919_S302_L001	PRNP	-9	-9
1	21093_S277_L001	PRNP	mh_6	mh_6
2	21094_S278_L001	PRNP	mh_1	mh_4
3	21095_S279_L001	PRNP	mh_4	mh_4
4	21098_S280_L001	PRNP	mh_3	mh_4
5	21099_S281_L001	PRNP	mh_3	mh_6
6	21101_S282_L001	PRNP	mh_4	mh_5
7	21102_S283_L001	PRNP	mh_1	mh_1
8	21104_S284_L001	PRNP	mh_4	mh_5
9	21105_S285_L001	PRNP	mh_0	mh_4
10	21106_S286_L001	PRNP	mh_6	mh_6
11	21107_S287_L001	PRNP	mh_4	mh_4
12	21110_S290_L001	PRNP	mh_3	mh_4
13	21111_S291_L001	PRNP	mh_1	mh_4
14	21112_S292_L001	PRNP	mh_6	mh_6
15	21113_S293_L001	PRNP	mh_6	mh_6
16	21115_S289_L001	PRNP	mh_1	mh_4
17	21118_S294_L001	PRNP	mh_6	mh_6
18	21297_S295_L001	PRNP	mh_2	mh_6
19	21298_S296_L001	PRNP	mh_4	mh_4
20	21299_S297_L001	PRNP	mh_3	mh_4
21	21301_S298_L001	PRNP	mh_4	mh_6
22	21303_S300_L001	PRNP	mh_3	mh_4
23	21304_S301_L001	PRNP	mh_2	mh_4
24	21305_S303_L001	PRNP	mh_2	mh_4
25	21306_S304_L001	PRNP	mh_1	mh_4
26	21307_S305_L001	PRNP	-9	-9
27	21309_S306_L001	PRNP	mh_4	mh_4
28	22643_S336_L001	PRNP	-9	-9
29	22644_S337_L001	PRNP	-9	-9
30	22645_S338_L001	PRNP	mh_4	mh_4
31	22646_S339_L001	PRNP	-9	-9
32	22647_S340_L001	PRNP	mh_4	mh_4
33	22648_S341_L001	PRNP	mh_4	mh_6

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“-9” null allele

Step 4.2.2.3 MH figure (all samples)



Step 4.2.3.1 Comprehensive MP table (locus)



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Microtype Viewing

Locus: PRNP

microtype sequence | microhap table (single) | **micropep table (single)** | microhap table (all) | micropep table (all)

comprehensive | simple | figure

“-9” null allele

Against reference

id	sample	locus	allele	readt	read	rprop	mprop	zygosity	baseChange	mp_seq
0	20919_S302_L001	PRNP	mp_1	66	16	0.2424	0.5333	inconclusive	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
1	20919_S302_L001	PRNP	mp_2	66	14	0.2121	0.7778	inconclusive		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
2	21093_S277_L001	PRNP	mp_1	66	36	0.5455	0.9474	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
3	21093_S277_L001	PRNP	-9	66	2	0.0303	0.0	homo		
4	21094_S278_L001	PRNP	mp_4	230	50	0.2174	0.5102	heter	2(S G)	GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
5	21094_S278_L001	PRNP	mp_2	230	48	0.2087	0.0	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
6	21095_S279_L001	PRNP	mp_2	192	48	0.25	0.9231	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
7	21095_S279_L001	PRNP	-9	192	4	0.0208	0.0	homo		
8	21098_S280_L001	PRNP	mp_2	182	74	0.4066	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
9	21098_S280_L001	PRNP	mp_2	182	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
10	21099_S281_L001	PRNP	mp_2	160	60	0.375	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
11	21099_S281_L001	PRNP	mp_2	160	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
12	21101_S282_L001	PRNP	mp_2	72	20	0.2778	0.5556	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
13	21101_S282_L001	PRNP	mp_1	72	16	0.2222	0.0	heter	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
14	21102_S283_L001	PRNP	mp_4	192	80	0.4167	0.9524	homo		GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
15	21102_S283_L001	PRNP	-9	192	4	0.0208	0.0	homo		
16	21104_S284_L001	PRNP	mp_1	290	76	0.2621	0.6786	heter	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
17	21104_S284_L001	PRNP	mp_2	290	36	0.1241	0.0	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
18	21105_S285_L001	PRNP	mp_2	214	34	0.1589	0.5152	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
19	21105_S285_L001	PRNP	mp_3	214	32	0.1495	0.0	heter	2(S G)42(M V)	GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPMDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
20	21106_S286_L001	PRNP	mp_2	336	156	0.4643	0.975	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
21	21106_S286_L001	PRNP	-9	336	4	0.0119	0.0	homo		
22	21107_S287_L001	PRNP	mp_2	262	98	0.374	0.9608	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
23	21107_S287_L001	PRNP	-9	262	4	0.0153	0.0	homo		
24	21110_S290_L001	PRNP	mp_2	158	66	0.4177	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
25	21110_S290_L001	PRNP	mp_2	158	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
26	21111_S291_L001	PRNP	mp_2	132	28	0.2121	0.6087	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
27	21111_S291_L001	PRNP	mp_4	132	18	0.1364	0.0	heter	2(S G)	GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
28	21112_S292_L001	PRNP	mp_2	124	62	0.5	0.9688	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
29	21112_S292_L001	PRNP	-9	124	2	0.0161	0.0	homo		
30	21113_S293_L001	PRNP	mp_2	206	94	0.4563	0.9592	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
31	21113_S293_L001	PRNP	-9	206	4	0.0194	0.0	homo		
32	21115_S289_L001	PRNP	mp_4	220	62	0.2818	0.5254	heter	2(S G)	GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM
33	21115_S289_L001	PRNP	mp_2	220	56	0.2545	0.0	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKM

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Step 4.2.3.2 Simple MP table (locus)


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◇ Microtype Viewing

Locus: PRNP

microtype sequence microhap table (single) **micropep table (single)** microhap table (all) micropep table (all)

comprehensive **simple** figure

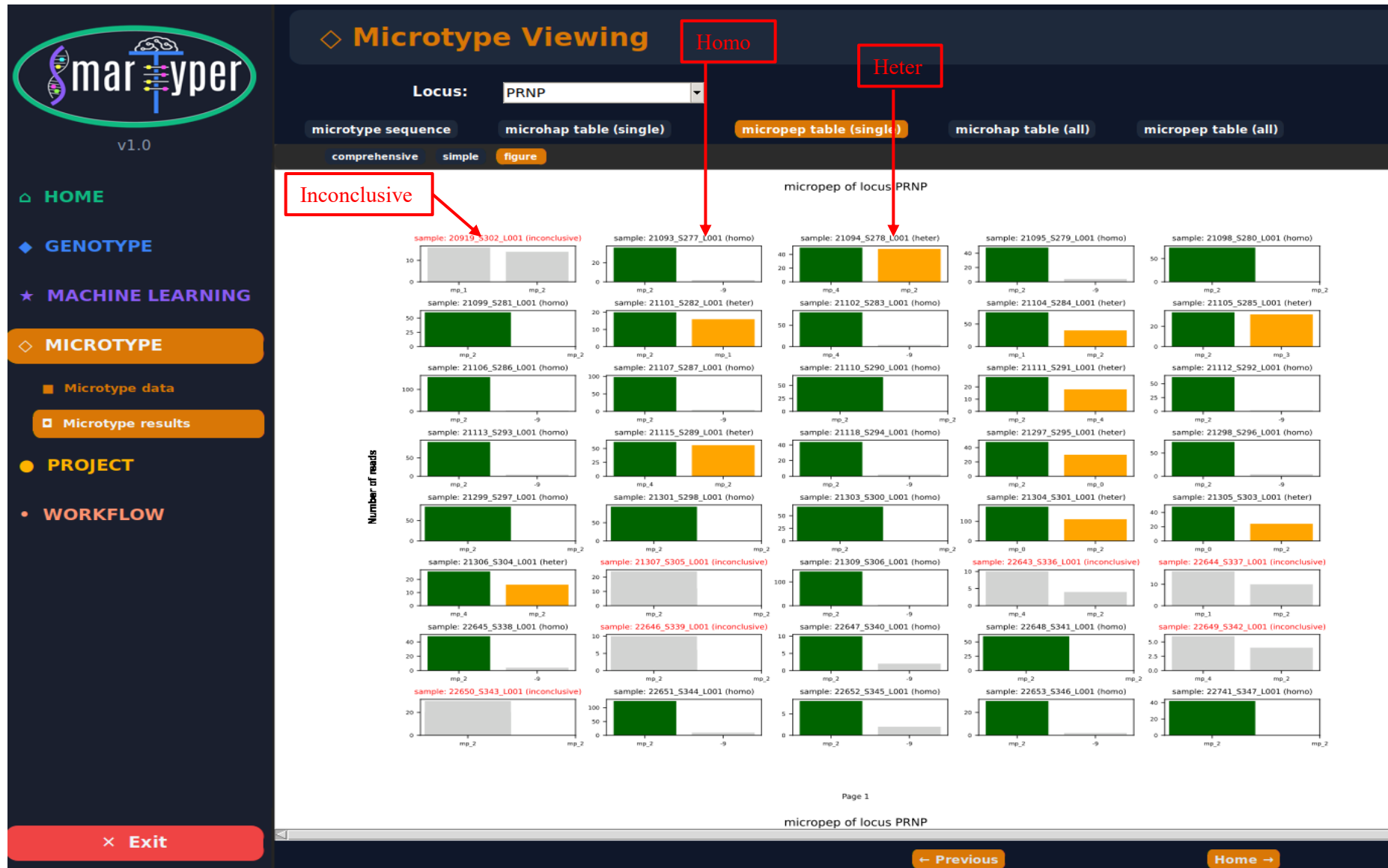
id	sample	locus	allele1	allele2
0	20919_S302_L001	PRNP	-9	-9
1	21093_S277_L001	PRNP	mp_2	mp_2
2	21094_S278_L001	PRNP	mp_4	mp_2
3	21095_S279_L001	PRNP	mp_2	mp_2
4	21098_S280_L001	PRNP	mp_2	mp_2
5	21099_S281_L001	PRNP	mp_2	mp_2
6	21101_S282_L001	PRNP	mp_2	mp_1
7	21102_S283_L001	PRNP	mp_4	mp_4
8	21104_S284_L001	PRNP	mp_1	mp_2
9	21105_S285_L001	PRNP	mp_2	mp_3
10	21106_S286_L001	PRNP	mp_2	mp_2
11	21107_S287_L001	PRNP	mp_2	mp_2
12	21110_S290_L001	PRNP	mp_2	mp_2
13	21111_S291_L001	PRNP	mp_2	mp_4
14	21112_S292_L001	PRNP	mp_2	mp_2
15	21113_S293_L001	PRNP	mp_2	mp_2
16	21115_S289_L001	PRNP	mp_4	mp_2
17	21118_S294_L001	PRNP	mp_2	mp_2
18	21297_S295_L001	PRNP	mp_2	mp_0
19	21298_S296_L001	PRNP	mp_2	mp_2
20	21299_S297_L001	PRNP	mp_2	mp_2
21	21301_S298_L001	PRNP	mp_2	mp_2
22	21303_S300_L001	PRNP	mp_2	mp_2
23	21304_S301_L001	PRNP	mp_0	mp_2
24	21305_S303_L001	PRNP	mp_0	mp_2
25	21306_S304_L001	PRNP	mp_4	mp_2
26	21307_S305_L001	PRNP	-9	-9
27	21309_S306_L001	PRNP	mp_2	mp_2
28	22643_S336_L001	PRNP	-9	-9
29	22644_S337_L001	PRNP	-9	-9
30	22645_S338_L001	PRNP	mp_2	mp_2
31	22646_S339_L001	PRNP	-9	-9
32	22647_S340_L001	PRNP	mp_2	mp_2
33	22648_S341_L001	PRNP	mp_2	mp_2

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“-9” null allele

Step 4.2.3.3 MP figure (all samples)



Step 4.2.4.1 Comprehensive MH table (all loci)



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Microtype Viewing

Locus: PRNP

microtype sequence
microhap table (single)
micropep table (single)
microhap table (all)
micropep table (all)

comprehensive
simple

Id	sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity	baseChange	
0	20919_S302_L001	PRNP	mh_5	66	16	0.2424	0.5333	0.7368	1 2	N	inconclusive	347(C T)	GG
1	20919_S302_L001	PRNP	mh_6	66	14	0.2121	0.7778	0.7368	1 2	N	inconclusive	60(C T)	GG
2	20919_S302_L001	PRNP2	mh_4	62	16	0.2581	0.5333	0.7059	1 2	N	inconclusive	334(C T)	
3	20919_S302_L001	PRNP2	mh_5	62	14	0.2258	0.7778	0.7059	1 2	N	inconclusive	47(C T)	
4	21093_S277_L001	PRNP	mh_6	66	36	0.5455	0.9474	0.9375	1 0	N	homo	60(C T)	GG
5	21093_S277_L001	PRNP	-9	66	2	0.0303	0.5	0.9375	0 0	N	homo	5(T C)60(C T)	GG
6	21093_S277_L001	PRNP2	mh_5	66	38	0.5758	0.95	0.9333	1 0	N	homo	47(C T)	
7	21093_S277_L001	PRNP2	-9	66	2	0.0303	0.5	0.9333	0 0	N	homo	47(C T)141(T C)	
8	21094_S278_L001	PRNP	mh_1	230	50	0.2174	0.5102	0.5294	1 1	N	heter	7(G A)	GG
9	21094_S278_L001	PRNP	mh_4	230	48	0.2087	0.96	0.5294	0 1	N	heter	nan	GG
10	21094_S278_L001	PRNP2	mh_3	226	98	0.4336	0.9608	0.9836	0 0	N	homo	nan	
11	21094_S278_L001	PRNP2	-9	226	4	0.0177	0.5	0.9836	0 0	N	homo	108(A G)	
12	21095_S279_L001	PRNP	mh_4	192	48	0.25	0.9231	0.9718	0 0	N	homo	nan	GG
13	21095_S279_L001	PRNP	-9	192	4	0.0208	0.5	0.9718	0 0	N	homo	279(C T)316(G A)	GG
14	21095_S279_L001	PRNP2	mh_3	186	54	0.2903	0.931	0.9846	0 0	N	homo	nan	
15	21095_S279_L001	PRNP2	-9	186	4	0.0215	0.5	0.9846	0 0	N	homo	266(C T)303(G A)	
16	21098_S280_L001	PRNP	mh_4	182	42	0.2308	0.5676	0.6296	0 1	N	heter	nan	GG
17	21098_S280_L001	PRNP	mh_3	182	32	0.1758	0.8889	0.6296	1 1	N	heter	156(T C)	GG
18	21098_S280_L001	PRNP2	mh_3	178	40	0.2247	0.5556	0.6226	0 1	N	heter	nan	
19	21098_S280_L001	PRNP2	mh_2	178	32	0.1798	0.8889	0.6226	1 1	N	heter	143(T C)	
20	21099_S281_L001	PRNP	mh_6	160	32	0.2	0.5333	0.5686	1 2	N	heter	60(C T)	GG
21	21099_S281_L001	PRNP	mh_3	160	28	0.175	0.875	0.5686	1 2	N	heter	156(T C)	GG
22	21099_S281_L001	PRNP2	mh_5	158	34	0.2152	0.5484	0.551	1 2	N	heter	47(C T)	
23	21099_S281_L001	PRNP2	mh_2	158	28	0.1772	0.875	0.551	1 2	N	heter	143(T C)	
24	21101_S282_L001	PRNP	mh_4	72	20	0.2778	0.5556	0.5	0 1	N	heter	nan	GG
25	21101_S282_L001	PRNP	mh_5	72	16	0.2222	0.8889	0.5	1 1	N	heter	347(C T)	GG
26	21101_S282_L001	PRNP2	mh_3	72	22	0.3056	0.5789	0.5263	0 1	N	heter	nan	
27	21101_S282_L001	PRNP2	mh_4	72	16	0.2222	0.8889	0.5263	1 1	N	heter	334(C T)	
28	21102_S283_L001	PRNP	mh_1	192	80	0.4167	0.9524	0.9818	1 0	N	homo	7(G A)	GG
29	21102_S283_L001	PRNP	-9	192	4	0.0208	0.5	0.9818	0 0	N	homo	7(G A)14(A G)	GG
30	21102_S283_L001	PRNP2	mh_3	190	84	0.4421	0.9545	0.9808	0 0	N	homo	nan	
31	21102_S283_L001	PRNP2	-9	190	4	0.0211	0.5	0.9808	0 0	N	homo	1(A G)	
32	21104_S284_L001	PRNP	mh_5	290	76	0.2621	0.6786	0.6067	1 1	N	heter	347(C T)	GG

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Microtype Viewing

Locus: PRNP

microtype sequence microhap table (single) micropep table (single) **microhap table (all)** micropep table (all)

comprehensive **simple**

Id	sample	PRNP2_allele1	PRNP2_allele2	PRNP_allele1	PRNP_allele2
0	20919_S302_L001	-9	-9	-9	-9
1	21093_S277_L001	mh_5	mh_5	mh_6	mh_6
2	21094_S278_L001	mh_3	mh_3	mh_1	mh_4
3	21095_S279_L001	mh_3	mh_3	mh_4	mh_4
4	21098_S280_L001	mh_2	mh_3	mh_3	mh_4
5	21099_S281_L001	mh_2	mh_5	mh_3	mh_6
6	21101_S282_L001	mh_3	mh_4	mh_4	mh_5
7	21102_S283_L001	mh_3	mh_3	mh_1	mh_1
8	21104_S284_L001	mh_3	mh_4	mh_4	mh_5
9	21105_S285_L001	mh_1	mh_3	mh_0	mh_4
10	21106_S286_L001	mh_5	mh_5	mh_6	mh_6
11	21107_S287_L001	mh_3	mh_3	mh_4	mh_4
12	21110_S290_L001	mh_2	mh_3	mh_3	mh_4
13	21111_S291_L001	mh_3	mh_3	mh_1	mh_4
14	21112_S292_L001	mh_5	mh_5	mh_6	mh_6
15	21113_S293_L001	mh_5	mh_5	mh_6	mh_6
16	21115_S289_L001	mh_3	mh_3	mh_1	mh_4
17	21118_S294_L001	mh_5	mh_5	mh_6	mh_6
18	21297_S295_L001	mh_0	mh_5	mh_2	mh_6
19	21298_S296_L001	mh_3	mh_3	mh_4	mh_4
20	21299_S297_L001	mh_2	mh_3	mh_3	mh_4
21	21301_S298_L001	mh_3	mh_5	mh_4	mh_6
22	21303_S300_L001	mh_2	mh_3	mh_3	mh_4
23	21304_S301_L001	mh_0	mh_3	mh_2	mh_4
24	21305_S303_L001	mh_0	mh_3	mh_2	mh_4
25	21306_S304_L001	mh_3	mh_3	mh_1	mh_4
26	21307_S305_L001	-9	-9	-9	-9
27	21309_S306_L001	mh_3	mh_3	mh_4	mh_4
28	22643_S336_L001	-9	-9	-9	-9
29	22644_S337_L001	-9	-9	-9	-9
30	22645_S338_L001	mh_3	mh_3	mh_4	mh_4
31	22646_S339_L001	-9	-9	-9	-9
32	22647_S340_L001	mh_3	mh_3	mh_4	mh_4

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Step 4.2.4.1 Comprehensive MP table (all loci)



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Microtype Viewing

Locus: PRNP

microtype sequence microhap table (single) micropep table (single) microhap table (all) **micropep table (all)**

comprehensive simple

Id	sample	locus	allele	readt	read	rprop	mprop	zygosity	baseChange	mp_seq
0	20919_S302_L001	PRNP	mp_1	66	16	0.2424	0.5333	inconclusive	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
1	20919_S302_L001	PRNP	mp_2	66	14	0.2121	0.7778	inconclusive		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
2	20919_S302_L001	PRNP2	mp_2	62	16	0.2581	0.5333	inconclusive	110(L P)	MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
3	20919_S302_L001	PRNP2	mp_3	62	14	0.2258	0.7778	inconclusive		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
4	21093_S277_L001	PRNP	mp_2	66	36	0.5455	0.9474	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
5	21093_S277_L001	PRNP	-9	66	2	0.0303	0.0	homo		
6	21093_S277_L001	PRNP2	mp_3	66	38	0.5758	0.95	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
7	21093_S277_L001	PRNP2	-9	66	2	0.0303	0.0	homo		
8	21094_S278_L001	PRNP	mp_4	230	50	0.2174	0.5102	heter	2(S G)	GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
9	21094_S278_L001	PRNP	mp_2	230	48	0.2087	0.0	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
10	21094_S278_L001	PRNP2	mp_3	226	98	0.4336	0.9608	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
11	21094_S278_L001	PRNP2	-9	226	4	0.0177	0.0	homo		
12	21095_S279_L001	PRNP	mp_2	192	48	0.25	0.9231	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
13	21095_S279_L001	PRNP	-9	192	4	0.0208	0.0	homo		
14	21095_S279_L001	PRNP2	mp_3	186	54	0.2903	0.931	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
15	21095_S279_L001	PRNP2	-9	186	4	0.0215	0.0	homo		
16	21098_S280_L001	PRNP	mp_2	182	74	0.4066	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
17	21098_S280_L001	PRNP	mp_2	182	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
18	21098_S280_L001	PRNP2	mp_3	178	72	0.4045	1.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
19	21098_S280_L001	PRNP2	mp_3	178	0	0.0	0.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
20	21099_S281_L001	PRNP	mp_2	160	60	0.375	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
21	21099_S281_L001	PRNP	mp_2	160	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
22	21099_S281_L001	PRNP2	mp_3	158	62	0.3924	1.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
23	21099_S281_L001	PRNP2	mp_3	158	0	0.0	0.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
24	21101_S282_L001	PRNP	mp_2	72	20	0.2778	0.5556	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
25	21101_S282_L001	PRNP	mp_1	72	16	0.2222	0.0	heter	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
26	21101_S282_L001	PRNP2	mp_3	72	22	0.3056	0.5789	heter		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
27	21101_S282_L001	PRNP2	mp_2	72	16	0.2222	0.0	heter	110(L P)	MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
28	21102_S283_L001	PRNP	mp_4	192	80	0.4167	0.9524	homo		GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
29	21102_S283_L001	PRNP	-9	192	4	0.0208	0.0	homo		
30	21102_S283_L001	PRNP2	mp_3	190	84	0.4421	0.9545	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI
31	21102_S283_L001	PRNP2	-9	190	4	0.0211	0.0	homo		
32	21104_S284_L001	PRNP	mp_1	290	76	0.2621	0.6786	heter	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYRPVDQYNNQNTFVHDCVNITVKQHTVTTTTKGENFTETDIKMMI

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Locus: PRNP

microtype sequence microhap table (single) micropep table (single) microhap table (all) **micropep table (all)**

comprehensive **simple**

Id	sample	PRNP_allele1	PRNP_allele2	PRNP2_allele1	PRNP2_allele2
0	20919_S302_L001	-9	-9	-9	-9
1	21093_S277_L001	mp_2	mp_2	mp_3	mp_3
2	21094_S278_L001	mp_4	mp_2	mp_3	mp_3
3	21095_S279_L001	mp_2	mp_2	mp_3	mp_3
4	21098_S280_L001	mp_2	mp_2	mp_3	mp_3
5	21099_S281_L001	mp_2	mp_2	mp_3	mp_3
6	21101_S282_L001	mp_2	mp_1	mp_3	mp_2
7	21102_S283_L001	mp_4	mp_4	mp_3	mp_3
8	21104_S284_L001	mp_1	mp_2	mp_2	mp_3
9	21105_S285_L001	mp_2	mp_3	mp_3	mp_1
10	21106_S286_L001	mp_2	mp_2	mp_3	mp_3
11	21107_S287_L001	mp_2	mp_2	mp_3	mp_3
12	21110_S290_L001	mp_2	mp_2	mp_3	mp_3
13	21111_S291_L001	mp_2	mp_4	mp_3	mp_3
14	21112_S292_L001	mp_2	mp_2	mp_3	mp_3
15	21113_S293_L001	mp_2	mp_2	mp_3	mp_3
16	21115_S289_L001	mp_4	mp_2	mp_3	mp_3
17	21118_S294_L001	mp_2	mp_2	mp_3	mp_3
18	21297_S295_L001	mp_2	mp_0	mp_3	mp_0
19	21298_S296_L001	mp_2	mp_2	mp_3	mp_3
20	21299_S297_L001	mp_2	mp_2	mp_3	mp_3
21	21301_S298_L001	mp_2	mp_2	mp_3	mp_3
22	21303_S300_L001	mp_2	mp_2	mp_3	mp_3
23	21304_S301_L001	mp_0	mp_2	mp_0	mp_3
24	21305_S303_L001	mp_0	mp_2	mp_0	mp_3
25	21306_S304_L001	mp_4	mp_2	mp_3	mp_3
26	21307_S305_L001	-9	-9	-9	-9
27	21309_S306_L001	mp_2	mp_2	mp_3	mp_3
28	22643_S336_L001	-9	-9	-9	-9
29	22644_S337_L001	-9	-9	-9	-9
30	22645_S338_L001	mp_2	mp_2	mp_3	mp_3
31	22646_S339_L001	-9	-9	-9	-9
32	22647_S340_L001	mp_2	mp_2	mp_3	mp_3

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Step 4.2.5.1 Important files for microhaplotypes

All_cur_mh_table.txt

sample	locus	allele	readt	read	rprop	mprop	sprop	mut	indel	zygosity	baseChange	mh_seq	id
20919_S302_L001	PRNP	mh_5	66	16	0.2424	0.5333	0.7368	1 2	N	inconclusive	347(C T)	GGGCCTCGGTGGCTACA	
20919_S302_L001	PRNP	mh_6	66	14	0.2121	0.7778	0.7368	1 2	N	inconclusive	60(C T)	GGGCCTCGGTGGCTACATGCTGGGA	
20919_S302_L001	PRNP2	mh_4	62	16	0.2581	0.5333	0.7059	1 2	N	inconclusive	334(C T)	TACATGCTGGGAAGTGC	
20919_S302_L001	PRNP2	mh_5	62	14	0.2258	0.7778	0.7059	1 2	N	inconclusive	47(C T)	TACATGCTGGGAAGTGCCATGAGCA	
21093_S277_L001	PRNP	mh_6	66	36	0.5455	0.9474	0.9375	1 0	N	homo	60(C T)	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCAT	
21093_S277_L001	PRNP	-9	66	2	0.0303	0.5	0.9375	0 0	N	homo	5(T C)60(C T)	GGGCCCCGGTGGCTACATGCTGGGA	
21093_S277_L001	PRNP2	mh_5	66	38	0.5758	0.95	0.9333	1 0	N	homo	47(C T)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21093_S277_L001	PRNP2	-9	66	2	0.0303	0.5	0.9333	0 0	N	homo	47(C T)141(T C)	TACATGCTGGGAAGTGCCATGAGCA	
21094_S278_L001	PRNP	mh_1	230	50	0.2174	0.5102	0.5294	1 1	N	heter	7(G A)	GGGCCTCAGTGGCTACATGCTGGGAAGTGCCAT	
21094_S278_L001	PRNP	mh_4	230	48	0.2087	0.96	0.5294	0 1	N	heter		GGGCCTCGGTGGCTACATGCTGGGAAGTGCCAT	
21094_S278_L001	PRNP2	mh_3	226	98	0.4336	0.9608	0.9836	0 0	N	homo		TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21094_S278_L001	PRNP2	-9	226	4	0.0177	0.5	0.9836	0 0	N	homo	108(A G)	TACATGCTGGGAAGTGCCATGAGCA	
21095_S279_L001	PRNP	mh_4	192	48	0.25	0.9231	0.9718	0 0	N	homo		GGGCCTCGGTGGCTACATGCTGGGAAGTGCCAT	
21095_S279_L001	PRNP	-9	192	4	0.0208	0.5	0.9718	0 0	N	homo	279(C T)316(G A)	GGGCCTCGGTGGCTACA	
21095_S279_L001	PRNP2	mh_3	186	54	0.2903	0.931	0.9846	0 0	N	homo		TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21095_S279_L001	PRNP2	-9	186	4	0.0215	0.5	0.9846	0 0	N	homo	266(C T)303(G A)	TACATGCTGGGAAGTGC	
21098_S280_L001	PRNP	mh_4	182	42	0.2308	0.5676	0.6296	0 1	N	heter		GGGCCTCGGTGGCTACATGCTGGGAAGTGCCAT	
21098_S280_L001	PRNP	mh_3	182	32	0.1758	0.8889	0.6296	1 1	N	heter	156(T C)	GGGCCTCGGTGGCTACATGCTGGGA	
21098_S280_L001	PRNP2	mh_3	178	40	0.2247	0.5556	0.6226	0 1	N	heter		TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21098_S280_L001	PRNP2	mh_2	178	32	0.1798	0.8889	0.6226	1 1	N	heter	143(T C)	TACATGCTGGGAAGTGCCATGAGCA	
21099_S281_L001	PRNP	mh_6	160	32	0.2	0.5333	0.5686	1 2	N	heter	60(C T)	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCAT	
21099_S281_L001	PRNP	mh_3	160	28	0.175	0.875	0.5686	1 2	N	heter	156(T C)	GGGCCTCGGTGGCTACATGCTGGGA	
21099_S281_L001	PRNP2	mh_5	158	34	0.2152	0.5484	0.551	1 2	N	heter	47(C T)	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTT	
21099_S281_L001	PRNP2	mh_2	158	28	0.1772	0.875	0.551	1 2	N	heter	143(T C)	TACATGCTGGGAAGTGCCATGAGCA	
21101_S282_L001	PRNP	mh_4	72	20	0.2778	0.5556	0.5	0 1	N	heter		GGGCCTCGGTGGCTACATGCTGGGAAGTGCCAT	

All_cur_mh_sim_table.txt

sample	PRNP2_allele1	PRNP2_allele2	PRNP_allele1	PRNP_allele2
20919_S302_L001	-9	-9	-9	-9
21093_S277_L001	mh_5	mh_5	mh_6	mh_6
21094_S278_L001	mh_3	mh_3	mh_1	mh_4
21095_S279_L001	mh_3	mh_3	mh_4	mh_4
21098_S280_L001	mh_2	mh_3	mh_3	mh_4
21099_S281_L001	mh_2	mh_5	mh_3	mh_6
21101_S282_L001	mh_3	mh_4	mh_4	mh_5
21102_S283_L001	mh_3	mh_3	mh_1	mh_1
21104_S284_L001	mh_3	mh_4	mh_4	mh_5
21105_S285_L001	mh_1	mh_3	mh_0	mh_4
21106_S286_L001	mh_5	mh_5	mh_6	mh_6
21107_S287_L001	mh_3	mh_3	mh_4	mh_4
21110_S290_L001	mh_2	mh_3	mh_3	mh_4
21111_S291_L001	mh_3	mh_3	mh_1	mh_4
21112_S292_L001	mh_5	mh_5	mh_6	mh_6
21113_S293_L001	mh_5	mh_5	mh_6	mh_6
21115_S289_L001	mh_3	mh_3	mh_1	mh_4
21118_S294_L001	mh_5	mh_5	mh_6	mh_6
21297_S295_L001	mh_0	mh_5	mh_2	mh_6
21298_S296_L001	mh_3	mh_3	mh_4	mh_4
21299_S297_L001	mh_2	mh_3	mh_3	mh_4
21301_S298_L001	mh_3	mh_5	mh_4	mh_6
21303_S300_L001	mh_2	mh_3	mh_3	mh_4
21304_S301_L001	mh_0	mh_3	mh_2	mh_4
21305_S303_L001	mh_0	mh_3	mh_2	mh_4
21306_S304_L001	mh_3	mh_3	mh_1	mh_4
21307_S305_L001	-9	-9	-9	-9
21309_S306_L001	mh_3	mh_3	mh_4	mh_4
22643_S336_L001	-9	-9	-9	-9

Step 4.2.5.2 Important files for micropeptides

All_cur_mp_table.txt

sample	locus	allele	readt	read	rprop	mprop	zygosity	baseChange	mp_seq	mp_label
20919_S302_L001	PRNP	mp_1	66	16	0.2424	0.5333	inconclusive	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
20919_S302_L001	PRNP	mp_2	66	14	0.2121	0.7778	inconclusive		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
20919_S302_L001	PRNP2	mp_2	62	16	0.2581	0.5333	inconclusive	110(L P)	MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
20919_S302_L001	PRNP2	mp_3	62	14	0.2258	0.7778	inconclusive		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21093_S277_L001	PRNP	mp_2	66	36	0.5455	0.9474	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21093_S277_L001	PRNP	-9	66	2	0.0303	0.0	homo		-9	
21093_S277_L001	PRNP2	mp_3	66	38	0.5758	0.95	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21093_S277_L001	PRNP2	-9	66	2	0.0303	0.0	homo		-9	
21094_S278_L001	PRNP	mp_4	230	50	0.2174	0.5102	heter	2(S G)	GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21094_S278_L001	PRNP	mp_2	230	48	0.2087	0.0	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21094_S278_L001	PRNP2	mp_3	226	98	0.4336	0.9608	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21094_S278_L001	PRNP2	-9	226	4	0.0177	0.0	homo		-9	
21095_S279_L001	PRNP	mp_2	192	48	0.25	0.9231	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21095_S279_L001	PRNP	-9	192	4	0.0208	0.0	homo		-9	
21095_S279_L001	PRNP2	mp_3	186	54	0.2903	0.931	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21095_S279_L001	PRNP2	-9	186	4	0.0215	0.0	homo		-9	
21098_S280_L001	PRNP	mp_2	182	74	0.4066	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21098_S280_L001	PRNP	mp_2	182	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21098_S280_L001	PRNP2	mp_3	178	72	0.4045	1.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21098_S280_L001	PRNP2	mp_3	178	0	0.0	0.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21099_S281_L001	PRNP	mp_2	160	60	0.375	1.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21099_S281_L001	PRNP	mp_2	160	0	0.0	0.0	homo		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21099_S281_L001	PRNP2	mp_3	158	62	0.3924	1.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21099_S281_L001	PRNP2	mp_3	158	0	0.0	0.0	homo		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21101_S282_L001	PRNP	mp_2	72	20	0.2778	0.5556	heter		GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21101_S282_L001	PRNP	mp_1	72	16	0.2222	0.0	heter	115(L P)	GLGGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21101_S282_L001	PRNP2	mp_3	72	22	0.3056	0.5789	heter		MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21101_S282_L001	PRNP2	mp_2	72	16	0.2222	0.0	heter	110(L P)	MLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	
21102_S283_L001	PRNP	mp_4	192	80	0.4167	0.9524	homo		GLSGYMLGSAMSRPLIHFGNDYEDRYRENMYRPNQVYYR	

All_cur_mp_sim_table.txt

sample	PRNP_allele1	PRNP_allele2	PRNP2_allele1	PRNP2_allele2
20919_S302_L001	-9	-9	-9	-9
21093_S277_L001	mp_2	mp_2	mp_3	mp_3
21094_S278_L001	mp_4	mp_2	mp_3	mp_3
21095_S279_L001	mp_2	mp_2	mp_3	mp_3
21098_S280_L001	mp_2	mp_2	mp_3	mp_3
21099_S281_L001	mp_2	mp_2	mp_3	mp_3
21101_S282_L001	mp_2	mp_1	mp_3	mp_2
21102_S283_L001	mp_4	mp_4	mp_3	mp_3
21104_S284_L001	mp_1	mp_2	mp_2	mp_3
21105_S285_L001	mp_2	mp_3	mp_3	mp_1
21106_S286_L001	mp_2	mp_2	mp_3	mp_3
21107_S287_L001	mp_2	mp_2	mp_3	mp_3
21110_S290_L001	mp_2	mp_2	mp_3	mp_3
21111_S291_L001	mp_2	mp_4	mp_3	mp_3
21112_S292_L001	mp_2	mp_2	mp_3	mp_3
21113_S293_L001	mp_2	mp_2	mp_3	mp_3
21115_S289_L001	mp_4	mp_2	mp_3	mp_3
21118_S294_L001	mp_2	mp_2	mp_3	mp_3
21297_S295_L001	mp_2	mp_0	mp_3	mp_0
21298_S296_L001	mp_2	mp_2	mp_3	mp_3
21299_S297_L001	mp_2	mp_2	mp_3	mp_3
21301_S298_L001	mp_2	mp_2	mp_3	mp_3
21303_S300_L001	mp_2	mp_2	mp_3	mp_3

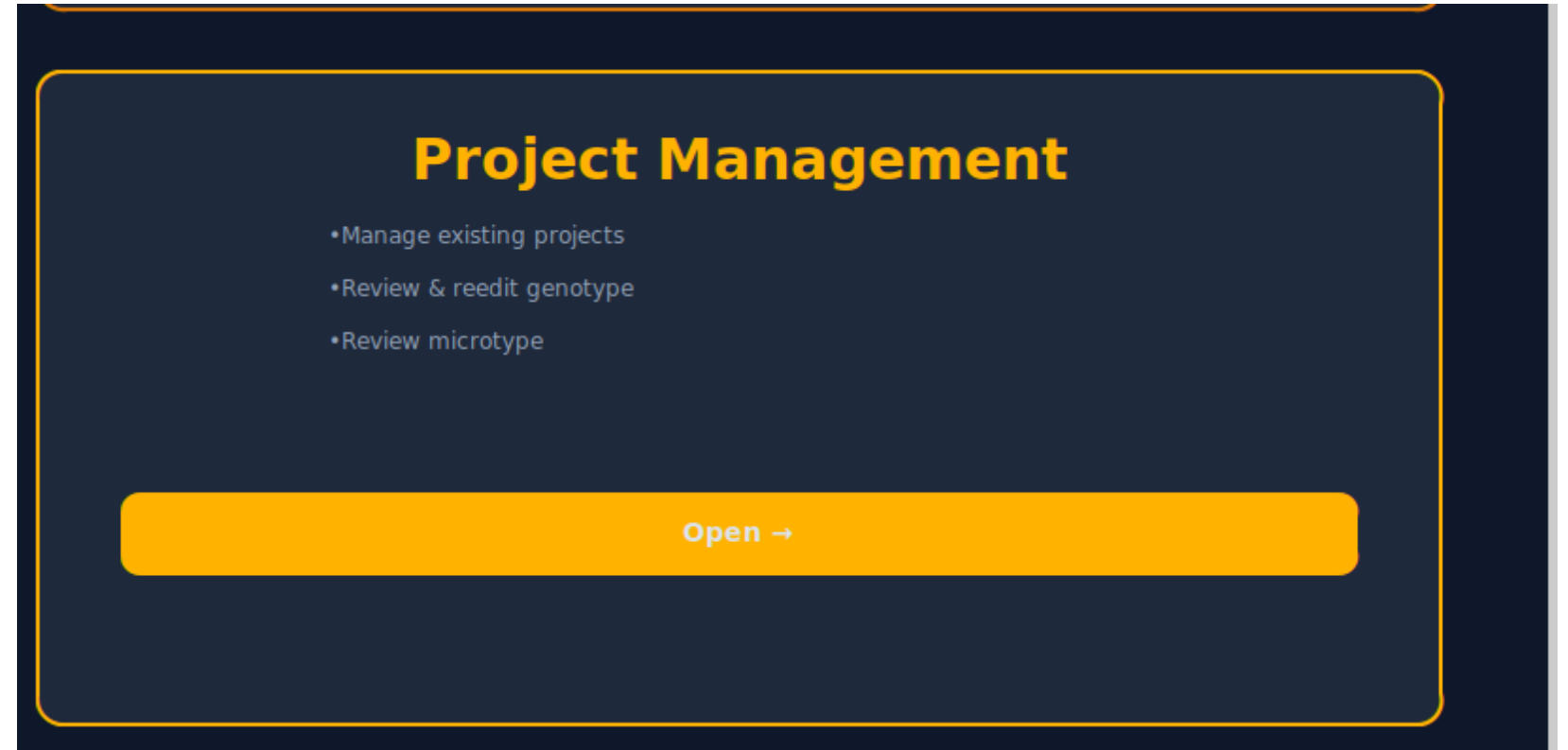
Step 4.2.5.3 Important file for microtyping

All_mt_lookup_table.txt

ocus	mh_label	mh_seq	mh_id	mp_label	mp_seq	mp_id
RNP	mh_0	GGGCCTCAGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP	mh_1	GGGCCTCAGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP	mh_2	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAACAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP	mh_3	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP	mh_4	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP	mh_5	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP	mh_6	GGGCCTCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAATGACTATGAGGACCGTTACTATCGTGAAAAACATGTA				
RNP2	mh_0	TACATGCTGGGAAGTGCCATGAACAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTACCGTTACCCCAAC				
RNP2	mh_1	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTACCGTTACCCCAAC				
RNP2	mh_2	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTACCGTTACCCCAAC				
RNP2	mh_3	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTACCGTTACCCCAAC				
RNP2	mh_4	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAACGACTATGAGGACCGTTACTATCGTGAAAAACATGTACCGTTACCCCAAC				
RNP2	mh_5	TACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAATGACTATGAGGACCGTTACTATCGTGAAAAACATGTACCGTTACCCCAAC				

Step 5 Project management

This module is designed to review existing genotype and microtype projects and re-editing genotype without re-running whole project



Step 5.1 Project data loading

The screenshot shows the marTyper v1.0 web interface. On the left is a sidebar with navigation links: HOME, GENOTYPE, MACHINE LEARNING, MICROTYPE, PROJECT (highlighted), WORKFLOW, and TUTORIAL. The main area is titled "Project Loading" and contains a "Num. thread:" input set to 10 and an "Input folder:" field with a "Browse" button. A "Confirm" button is next to the input folder field. A red callout box points to the "Confirm" button with the text: "2. Click 'Confirm' and wait for loading then click 'Next'".

A "Select Output Directory" dialog box is open, showing a file list. A red callout box points to the dialog with the text: "1. Must select the output folder of 'Genotype' or 'Microtype' modules containing 'project_genotype_session.dill' or 'project_microtype_session.dill' files". The dialog shows a directory path "/home/liup/project/data/prnp/newtest/out1" and a table of files.

Name	Type	Size	Modified
22768_S335_L001_sample_final_haplotype.txt	File	1.8 KB	2026-04-06 10:41
22768_S335_L001_sample_final_haplotype.txt	File	258.0 B	2026-04-06 10:41
22768_S335_L001_sample_final_haplotype.txt	File	1.0 KB	2026-04-06 10:41
22768_S335_L001_sample_final_haplotype.txt	File	21.2 KB	2026-04-06 10:41
22768_S335_L001_sample_final_haplotype.txt	File	1.8 KB	2026-04-06 10:40
22768_S335_L001_sample_final_haplotype.txt	File	258.0 B	2026-04-06 10:40
22768_S335_L001_sample_reads_distribution.pdf	File	14.3 KB	2026-04-06 10:40
22768_S335_L001_sample_snp.txt	File	1.0 KB	2026-04-06 10:40
All_sample_final_haplotype.txt	File	99.3 KB	2026-04-06 10:41
All_sample_final_ml.txt	File	9.1 KB	2026-04-06 10:41
All_sample_read_distribution.pdf	File	18.5 KB	2026-04-06 10:40
All_sample_read_quality.pdf	File	410.2 KB	2026-04-06 10:40
PRNP2_locus_genotype.pdf	File	27.4 KB	2026-04-06 10:41
PRNP_locus_genotype.pdf	File	27.5 KB	2026-04-06 10:41
PRNP_locus_reads_distribution.pdf	File	19.0 KB	2026-04-06 10:41
project_genotype_session.dill	File	12.2 MB	2026-04-06 10:41
smarter_log.txt	File	69.7 KB	2026-04-06 10:40

The dialog also has a "Selected:" field showing "out1" and buttons for "Select", "New Folder", and "Cancel". At the bottom of the main interface are "Previous" and "Next" buttons. A red arrow points from the "Next" button to the "Confirm" button in the main interface.

Step 5.2 To Result visualization (genotype output loaded)



v1.0

HOME

GENOTYPE

Data

Parameters

Run

Results

Results Output

MACHINE LEARNING

MICROTYPE

PROJECT

WORKFLOW

Exit

Result Viewing

Sample or Locus:

sample

locus

21093_S277_L001

Geno Combo

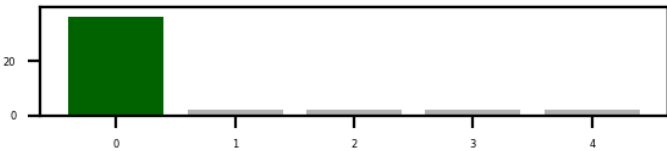
Geno Fig & Tab

All Sample Geno

Sample Reads

All Sample Reads

Sample: 21093_S277_L001, Marker: PRNP



Continue to review and edit genotype

id	sample	allele	baseChange	readt	read	rprop	mprop	sprop	mut	zygosity	indel	len
0	21093_S277_L001	PRNP_0	60(C T)	66	36	0.5455	0.9474	0.9375	1 0	homo	N	367
1	21093_S277_L001	PRNP_1	5(T C)60(C T)	66	2	0.0303	0.5	0.9375	0 0	homo	N	367
2	21093_S277_L001	PRNP_2	60(C T)154(T C)	66	2	0.0303	nan	nan	nan	inconclusive	nan	367
3	21093_S277_L001	PRNP_3	60(C T)162(T C)	66	2	0.0303	nan	nan	nan	inconclusive	nan	367
4	21093_S277_L001	PRNP_4	60(C T)248(T C)298(C T)	66	2	0.0303	nan	nan	nan	inconclusive	nan	367

***: GGGCCCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAAAGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTAACAGGCCAGTGGATCAGTATAATAACAGAACACCTTTGTGCAAGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAA

00: GGGCCCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAAAGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTAACAGGCCAGTGGATCAGTATAATAACAGAACACCTTTGTGCAAGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAA

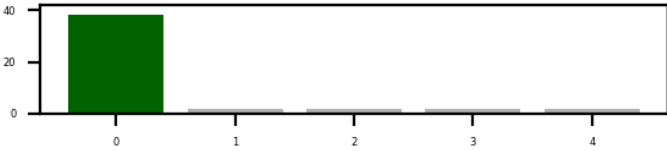
01: GGGCCCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAAAGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTAACAGGCCAGTGGATCAGTATAATAACAGAACACCTTTGTGCAAGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAA

02: GGGCCCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAAAGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTAACAGGCCAGTGGATCAGTATAATAACAGAACACCTTTGTGCAAGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAA

03: GGGCCCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAAAGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTAACAGGCCAGTGGATCAGTATAATAACAGAACACCTTTGTGCAAGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAA

04: GGGCCCGGTGGCTACATGCTGGGAAGTGCCATGAGCAGGCCTCTTATACATTTTGGCAAAGACTATGAGGACCGTTACTATCGTGAAAACATGTACCGTTACCCCAACCAAGTGTAACAGGCCAGTGGATCAGTATAATAACAGAACACCTTTGTGCAAGACTGTGTCAACATCACAGTCAAGCAACACACAGTCACCACCACCACCAAGGGGGAGAA

Sample: 21093_S277_L001, Marker: PRNP2



id	sample	allele	baseChange	readt	read	rprop	mprop	sprop	mut	zygosity	indel	len
0	21093_S277_L001	PRNP2_0	47(C T)	66	38	0.5758	0.95	0.9333	1 0	homo	N	338

Previous

Page 1 of 1

1

Go

Next

Previous

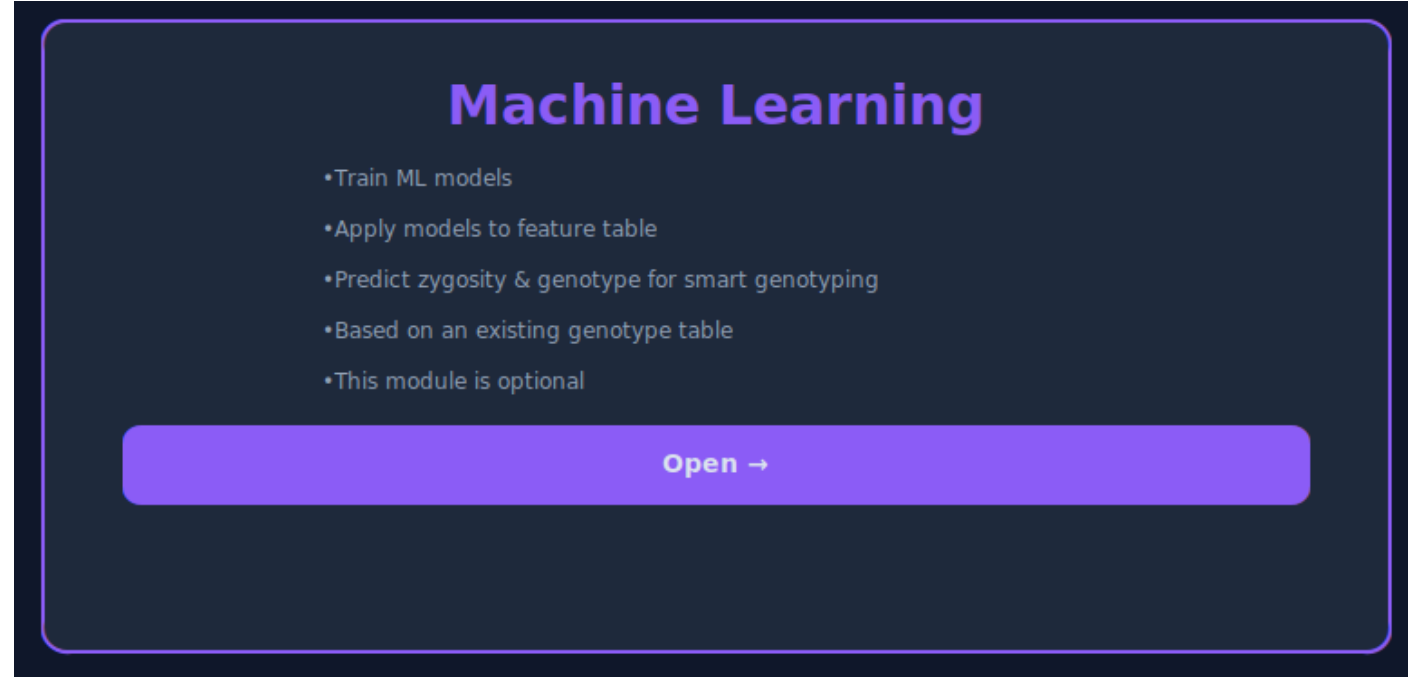
Next

Step 5.3 To Microtype visualization (microtype output loading)

[illegible]

Step 6 Machine learning

- This module is designed to train a supervised gradient-boosting model for each locus to increase the accuracy of automated genotyping and reduce manual genotype editing.
- Model training can be trained when needed with combination of all “All_sample_final_ml.txt” files

A dark-themed user interface for a 'Machine Learning' module. The title 'Machine Learning' is displayed in a large, bold, light blue font at the top. Below it, a list of five bullet points in a smaller, light blue font describes the module's functions: training ML models, applying models to a feature table, predicting zygosity and genotype for smart genotyping, being based on an existing genotype table, and being optional. At the bottom, there is a prominent, rounded rectangular button with a light blue gradient and the text 'Open →' in white.

Machine Learning

- Train ML models
- Apply models to feature table
- Predict zygosity & genotype for smart genotyping
- Based on an existing genotype table
- This module is optional

Open →

Step 6.1 Model Training

Model Training

Analysis type: ☒ Snp ☐ Sat

Training file:

Output folder:

Select the "All_sample_final_ml.txt" file from output of "Genotyping" module

Select or create an output folder and click "Confirm" wait for processing completed

Training output files are "All_training_models.pkl" and "All_loci_accuracy.txt"

Select File

Back Up /home/llup/project/data/prnp/newtest/out1

Name	Type	Size	Modified
22767_S334_L001_sample_final_ml.txt	File	254.0 B	2026-04-01 15:46
22767_S334_L001_sample_final_snp.txt	File	1.0 KB	2026-04-01 15:46
22767_S334_L001_sample_haplotype.txt	File	1.8 KB	2026-04-01 15:40
22767_S334_L001_sample_ml.txt	File	254.0 B	2026-04-01 15:40
22767_S334_L001_sample_snp.txt	File	1.0 KB	2026-04-01 15:40
22768_S335_L001_sample_amplicon.txt	File	45.6 KB	2026-04-01 15:40
22768_S335_L001_sample_error_rate.txt	File	2.1 KB	2026-04-01 15:40
22768_S335_L001_sample_final_haplotype.txt	File	1.8 KB	2026-04-01 15:46
22768_S335_L001_sample_final_ml.txt	File	258.0 B	2026-04-01 15:46
22768_S335_L001_sample_final_snp.txt	File	1.0 KB	2026-04-01 15:46
22768_S335_L001_sample_haplotype.txt	File	1.8 KB	2026-04-01 15:40
22768_S335_L001_sample_ml.txt	File	258.0 B	2026-04-01 15:40
22768_S335_L001_sample_snp.txt	File	1.0 KB	2026-04-01 15:40
All_sample_final_haplotype.txt	File	99.3 KB	2026-04-01 15:46
All_sample_final_ml.txt	File	9.1 KB	2026-04-01 15:46
project_genotype_session.dill	File	12.2 MB	2026-04-01 15:46
smartyper_log.txt	File	69.6 KB	2026-04-01 15:41

Selected: All_sample_final_ml.txt

Open Cancel

Success

Model training completed successfully!

OK

Training output files are "All_training_models.pkl" and "All_loci_accuracy.txt"

Exit Home

Step 6.2 Loading training model for a project

The screenshot displays the 'mar typer v1.0' web application interface. On the left is a sidebar with navigation links: HOME, GENOTYPE (selected), Data, Parameters, Run, Results, Results Output, MACHINE LEARNING, MICROTYPE, PROJECT, and WORKFLOW. At the bottom of the sidebar is an 'Exit' button.

The main panel is titled 'Data Loading' and contains the following fields:

- Analysis type: ☒ Snp ☐ Sat
- Read type: ☒ PE ☐ SE
- Input folder:
- Sample file:
- Loci file: ☐ reverse complement
- Sex file (optional): ☐ reverse complement
- training model (optional): (highlighted with a red box and an arrow pointing to the file selection dialog)
- Output folder:

A 'Confirm' button is located at the bottom right of the 'Data Loading' section.

A 'Select File' dialog box is open, showing the directory '/home/liup/project/data/prnp/newtest/ml1'. It contains a table with the following data:

Name	Type	Size	Modified
All_loci_accuracy.txt	File	119.0 B	2026-04-01 16:05
All_training_models.pkl	File	576.0 KB	2026-04-01 16:05

The file 'All_training_models.pkl' is selected. Below the table, the 'Selected:' field shows 'All_training_models.pkl'. At the bottom of the dialog are 'Open' and 'Cancel' buttons.

THE END!